MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

The pressure and density of a given mass of a diatomic gas $\left(\gamma = \frac{7}{5}\right)$ change adiabatically from (P, d) to (P', d'). If $\frac{d'}{d} = 32$, then $\frac{P'}{P}$ is ($\gamma =$ ratio of specific heats)

(A) $\frac{1}{128}$

(B) $\frac{1}{64}$

(C) 64

(D) 128

Q.2

A screw gauge gives the following reading when used to measure the diameter of a wire.

Main scale reading : 0 mm

Circular scale reading: 52 divisions

Given that 1 mm on the main scale corresponds to

100 divisions of the circular scale.

The diameter of the wire from the above data is

(A) 0.052 cm

(B) 0.026 cm

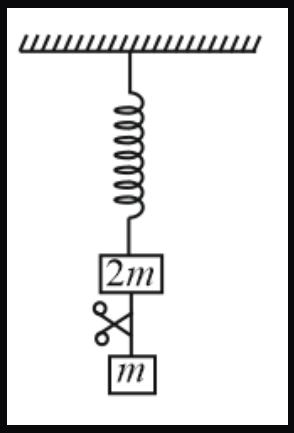
(C) 0.005 cm

(D) 0.52 cm

Q.3

System shown in the figure is in equilibrium and at rest. The spring and string are massless, now the string is cut. The

acceleration of mass $2m$ and m just after the string is cut will be



(A) $\frac{g}{2}$ upwards, g downwards

(B) g upwards, $\frac{g}{2}$ downwards

(C) g upwards, $2g$ downwards

(D) $2g$ upwards, g downwards

Q.4

One end of wire 2 m long and diameter 2 mm is fixed in a ceiling. A naughty boy of mass 10 kg jumps to catch the free end and stays there. The change in length of wire is [Take $g = 10\ \text{m s}^{-2}$, $Y = 2 \times 10^{11}\ \text{N m}^{-2}$]

(A) 31.85×10^{-5} m

(B) 2 mm

(C) 3 mm

(D) 4 m

Q.5

A pulse of light of duration 100 ns is absorbed completely by a small object initially at rest. Power of the pulse is 30 mW and the speed of light is 3 $\times 10^8$ ms^{-1}. The final momentum of the object is

(A) $0.3 \times 10^{-17}\ \text{kg ms}^{-1}$

(B) $1.0 \times 10^{-17}\ \text{kg ms}^{-1}$

(C) $3.0 \times 10^{-17}\ \text{kg ms}^{-1}$

(D) $9.0 \times 10^{-17}\ \text{kg ms}^{-1}$

Q.6

A body, starting from rest, is moved along a straight line by a machine delivering constant power. The distance moved by the body in time t is proportional to

(A) $t^{1/2}$

(B) $t^{3/4}$

(C) $t^{3/2}$

(D) $t^{1/4}$

Q.7

Assertion (A) When a wire of aluminium and another wire of silicon are heated from room temperature to $80^{\circ}\mathrm{C}$, the conductivity of aluminium increases and that of silicon decreases.
Reason (R) Aluminium has positive temperature coefficient of resistivity and silicon has negative temperature coefficient of resistivity.

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A).

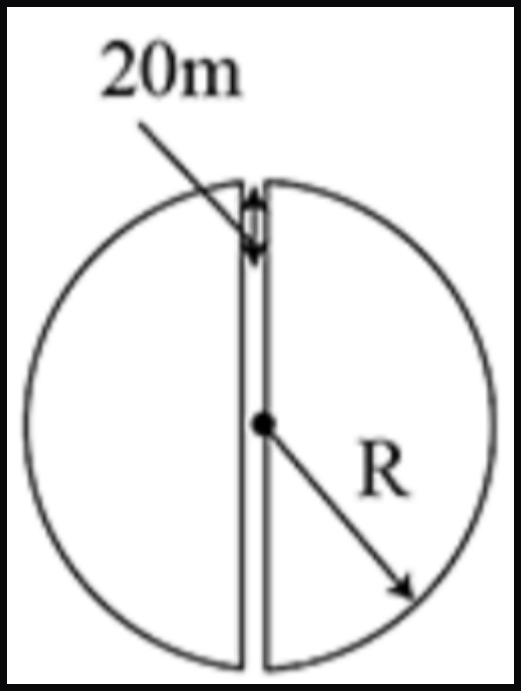
(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(C) (A) is correct but (R) is not correct.

(D) (A) is not correct but (R) is correct.

Q.8

A particle is dropped from depth of $20\ \mathrm{m}$ below the earth's surface along a tunnel created inside the planet of mass M with radius R as shown in the figure, the time period of oscillation is : (If $GM = 10R^3$, given $\pi^2 = 10$)



(A) $4s$

(B) $20\ \mathrm{s}$

(C) $2s$

(D) $10s$

Q.9

A body is travelling with $10\ \mathrm{ms}^{-1}$ on a rough horizontal surface. It's velocity after $2\ \mathrm{s}$ is $4\ \mathrm{ms}^{-1}$. The coefficient of kinetic friction between the block and the plane is
(acceleration due to gravity $= 10\ \mathrm{ms}^{-2}$)

(A) 0.4

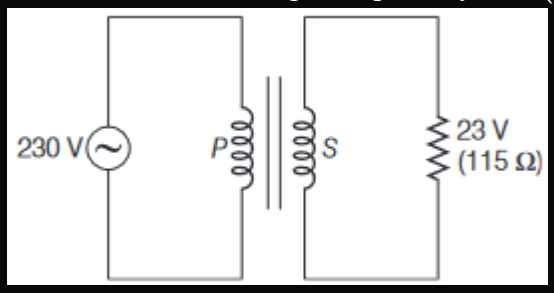
(B) 0.3

(C) 0.5

(D) 0.2

Q.10

Find the current through the primary coil (P) of the transformer shown below.



(A) 0.08 A

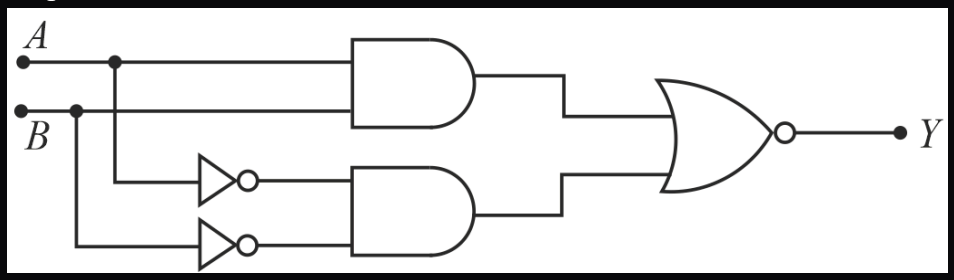
(B) 0.04 A

(C) 0.02 A

(D) 0.01 A

Q.11

In the given circuit, the binary inputs at A and B are both 1 in one case and both 0 in the next case. The respective outputs at Y in these two cases will be:



(A) $1, 1$

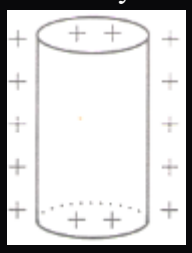
(B) $0, 0$

(C) $0, 1$

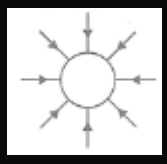
(D) $1, 0$

Q.12

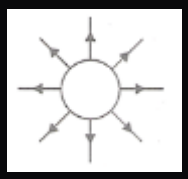
Which of the following figures correctly shows the top view sketch of the electric field lines for a uniformly charged hollow cylinder as shown in figure?



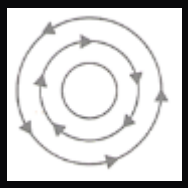
(A)



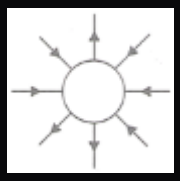
(B)



(C)



(D)



Q.13

The pass-axes of two polarisers were kept such that the incident unpolarised beam of intensity I_0, gets completely blocked. Another polariser was introduced in between these two polarisers with its pass-axis 60° with respect to the pass-axis of the first one. The output intensity would then become

(A) 0

(B) $\frac{3}{32}I_0$

(C) $\frac{3}{16}I_0$

(D) $\frac{3}{8}I_0$

Q.14

Statement (A) Two artificial satellites revolving in the same circular orbit have same period of revolution.
Statement (B) The orbital velocity is inversely proportional to the square root of radius of the orbit.
Statement (C) The escape velocity of the body is independent of the altitude of the point of projection.

(A) A, B, C are true

(B) A, B true C false

(C) A, C true B false

(D) B, C true A false

Q.15

In the following a statement of Assertion is followed by a statement of Reason.
Assertion: The mass defect involved in a chemical reaction is almost a million times smaller than that in a nuclear

reaction.
Reason: The mass energy interconversion does not take place in a chemical reaction.

(A) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion.

(B) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion.

(C) Assertion is true statement but Reason is false.

(D) Both Assertion and Reason are false statements.

Q.16

An organ pipe P_1, closed at one end and containing a gas of density ρ_1 is vibrating in its first harmonic. Another organ pipe P_2, open at both ends and containing a gas of density ρ_2 is vibrating in its third harmonic. Both the pipes are in resonance with a given tuning fork. If the compressibility of gases is equal in both pipes, the ratio of the lengths of P_1 and P_2 is (assume the given gases to be monoatomic)

(A) $\frac{1}{3}$

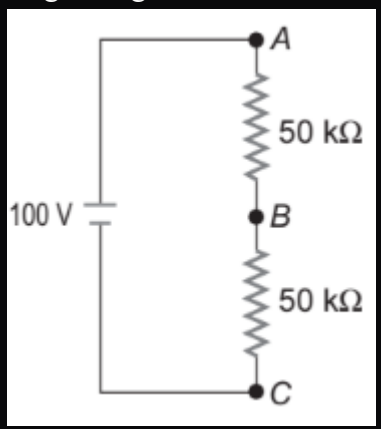
(B) 3

(C) $\frac{1}{6}\sqrt{\frac{\rho_1}{\rho_2}}$

(D) $\frac{1}{6}\sqrt{\frac{\rho_2}{\rho_1}}$

Q.17

In the adjacent shown circuit, a voltmeter of internal resistance R, when connected across B and C reads $\frac{100}{3}$ V. Neglecting the internal resistance of the battery, the value of R is



(A) $100\text{k}\Omega$

(B) $75\text{k}\Omega$

(C) $50\text{k}\Omega$

(D) $25\text{k}\Omega$

Q.18

The energy of gamma (γ) ray photon is E_γ and that of an X-ray photon is E_x. If the visible light photon has energy

of E_{v} , then we can say that

(A) $\mathrm{E_x} > \mathrm{E}_\gamma > \mathrm{E_v}$

(B) $\mathrm{E}_\gamma > \mathrm{E_v} > \mathrm{E_x}$

(C) $E_\gamma > E_{\mathrm{x}} > E_{\mathrm{v}}$

(D) $E_{\mathrm{x}} > E_{\mathrm{v}} > E_\gamma$

Q.19

A gas satisfies the relation $PV^{5/3} = K$ where P is pressure, V is volume and K is constant. The dimensions of constant K are

(A) $\mathrm{ML}^4\,\mathrm{T}^{-2}$

(B) $\mathrm{ML}^2\,\mathrm{T}^{-2}$

(C) $\mathrm{ML}^6\,\mathrm{T}^{-2}$

(D) MLT^{-2}

Q.20

Assertion (A) When an ideal gas is compressed adiabatically, its temperature and the average kinetic energy of the gas molecules increase.
Reason (R) The kinetic energy increases because of collisions of molecules with moving parts of wall only.

(A) A and R and true and R is correct explanation of A.

(B) A and R are true but R is not correct explanation of A.

(C) A is true and R is false.

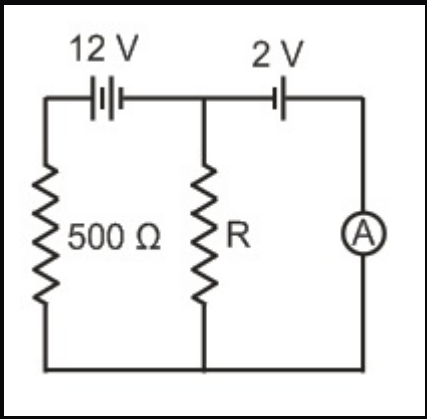
(D) A is false and R is true.

Q.21

In a series $L - C$ - R circuit, the inductive reactance is twice the resistance and the capacitive reactance is $1/3$ rd of the inductive reactance. The power factor of the circuit is K, find 10K.

Q.22

In the circuit shown below, the ammeter reading is zero. Then, the value of the resistance R is



Q.23

A hydrogen sample is prepared in a particular excited state A of quantum number, $n_A = 3$. The ground state energy of hydrogen atom is $-|E|$. The photons of energy $\frac{|E|}{12}$ are absorbed in the sample which results in the excitation of some electrons to excited state B of quantum number n_B, whose value is

Q.24

A particle of mass $0.6\ \mathrm{g}$ and having charge of $25\mathrm{nC}$ is moving horizontally with a uniform velocity $1.2 \times 10^4\ \mathrm{ms}^{-1}$ in a uniform magnetic field, then the value of the magnetic induction is $\left(g = 10\ \mathrm{ms}^{-2}\right)$

Q.25

An open cylindrical vessel filled with water is rotated together with water at angular speed $1\ \mathrm{rad\ s}^{-1}$. At this angular speed, the centre of the bottom is just exposed. If the angular speed is increased to $\sqrt{\frac{3}{2}}\ \mathrm{rad\ s}^{-1}$, how much area (in cm^2) of the bottom of the vessel is now exposed to surrounding? [Area of the cross-section of cylinder is $27\ \mathrm{cm}^2$]

Q.26

Match the following :

	List-I		List-II
(A)	Oxyhemoglobin	(i)	Hardness of water
(B)	EDTA	(ii)	Oxygen carrier
(C)	Hemoglobin	(iii)	Photosynthesis
(D)	Chlorophyll	(iv)	Oil of winter green
		(v)	Fe^{2+} paramagnetic

The correct match is

A B C D

(A) (v) (i) (ii) (iii)

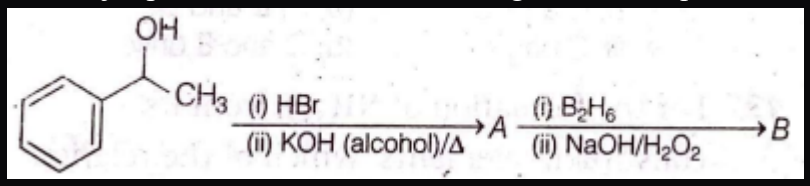
(B) (iv) (ii) (i) (iii)

(C) (iii) (i) (ii) (iv)

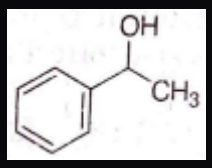
(D) (v) (ii) (iii) (i)

Q.27

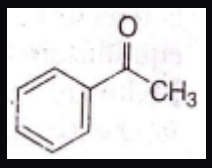
The major product ' B ' of the following reaction sequence is



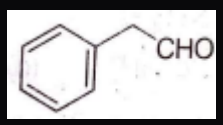
(A)



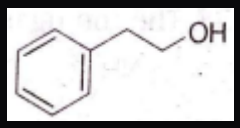
(B)



(C)



(D)



Q.28

Rusting of iron is catalysed by which of the following?

(A) Fe

(B) Zn

(C) $\mathrm{O_2}$

(D) $\mathrm{H^+}$

Q.29

A compound A contains C and H only and has molecular mass 72. It's photochlorination gives a mixture containing only one monochloro and two dichloro hydrocarbons. The IUPAC name is

(A) 2, 2-dimethylpropane

(B) n-pentane

(C) 2-methyl butane

(D) cyclopentane

Q.30

Which of the following is not a disproportionation reaction?

(A) $Hg_2Cl_2 \longrightarrow Hg + HgCl_2$

(B) $4H_3PO_3 \longrightarrow 3H_3PO_4 + PH_3$

(C) $Cl_2 + 2NaOH \longrightarrow NaCl + NaOCl + H_2O$

(D) $BaO_2 + H_2SP_4 \longrightarrow BaSO_4 + H_2O_2$

Q.31

The correct order of boiling points of $H_2O, H_2\ S, H_2\ Se$ and $H_2\ Te$ respectively is

(A) $H_2O > H_2\ S = H_2Se = H_2\ Te$

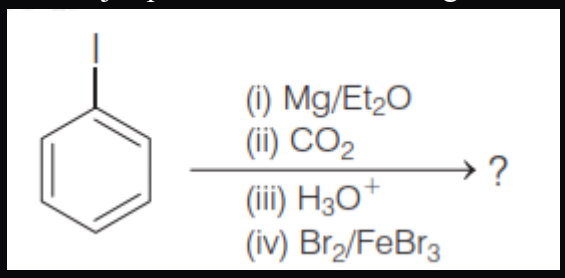
(B) $H_2O < H_2\ S < H_2Se < H_2\ Te$

(C) $H_2O > H_2\ S > H_2Se > H_2\ Te$

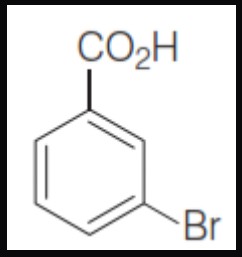
(D) $H_2O > H_2Te > H_2Se > H_2\ S$

Q.32

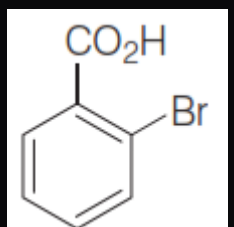
The major product of the following reaction sequence is



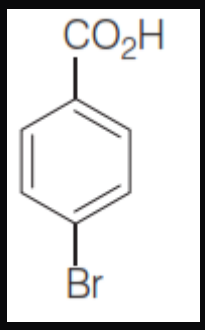
(A)



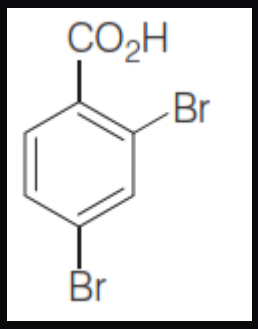
(B)



(C)

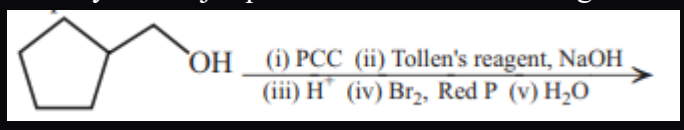


(D)

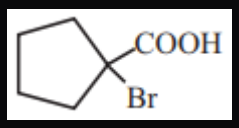


Q.33

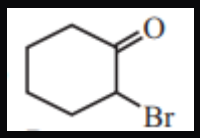
Identify the major product from the following reaction sequence



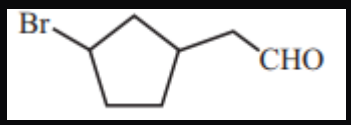
(A)



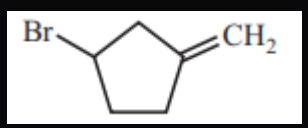
(B)



(C)

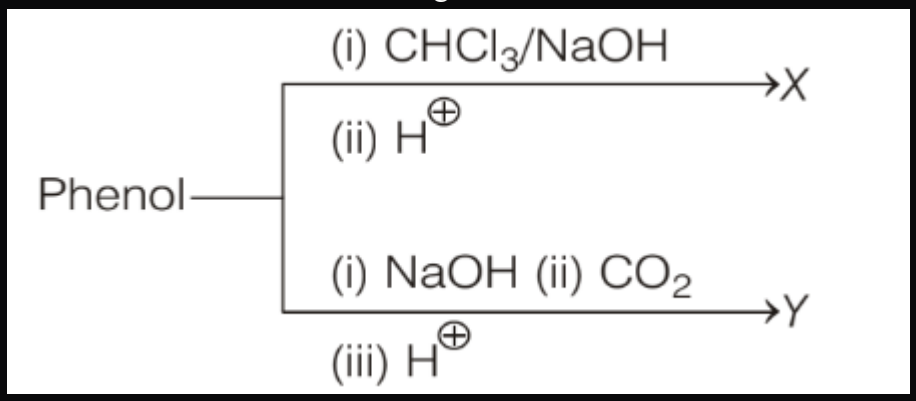


(D)

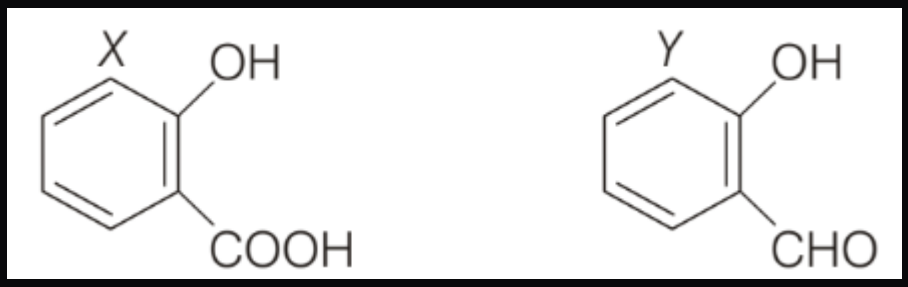


Q.34

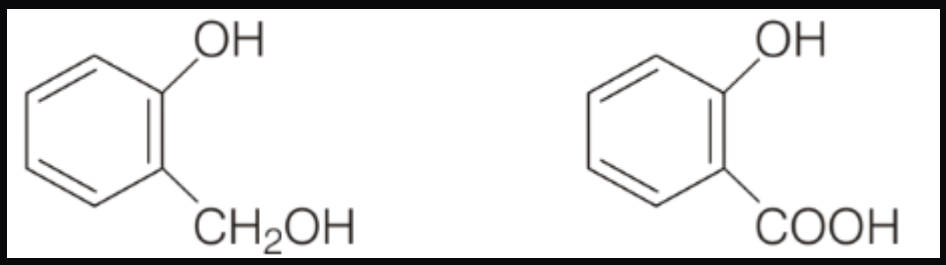
What are X and Y in the following reactions?



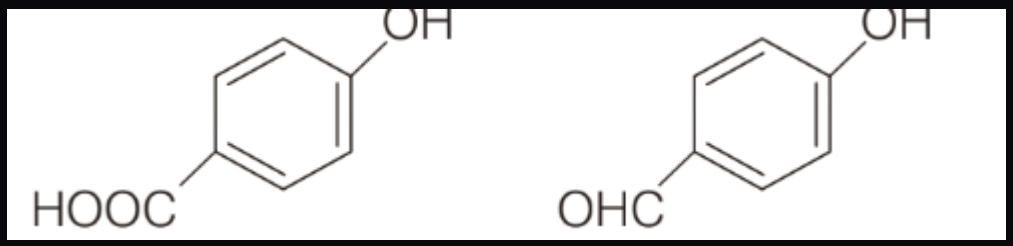
(A)



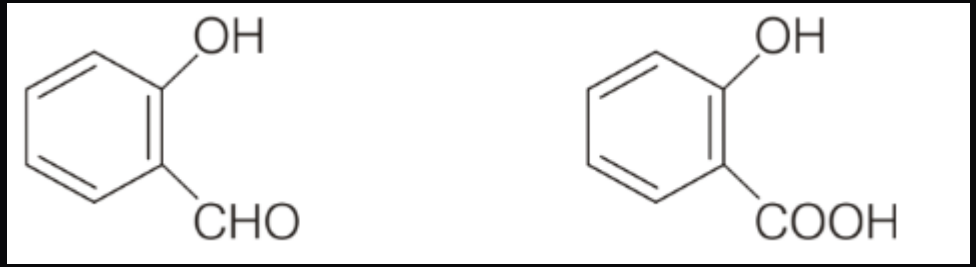
(B)



(C)



(D)



Q.35

Which of the following mixture form an ideal solution?

(A) $CCl_4 + C_7H_8$

(B) $CHCl_3 + C_6H_6$

(C) $H_2O + CH_3OH$

(D) $n - C_6H_{14} + n - C_7H_{16}$

Q.36

Choose the correct option regarding the following.
Assertion : Actinides exhibit a larger number of oxidation states than lanthanides.
Reason : There is a large energy gap between $5f, 6d$ and $7s$ sub-shells.

(A) Assertion and Reason are correct, and Reason is the correct explanation for the Assertion.

(B) Assertion and Reason are correct, and Reason is not the correct explanation for the Assertion.

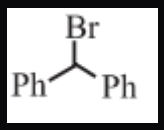
(C) Assertion is correct, Reason is wrong.

(D) Assertion is incorrect, Reason is correct.

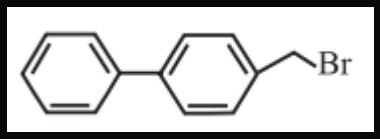
Q.37

The rate S_N1 reaction will be faster for which of the following bromides?

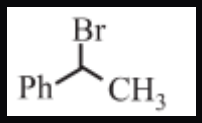
(A)



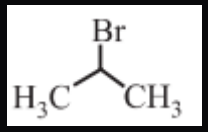
(B)



(C)

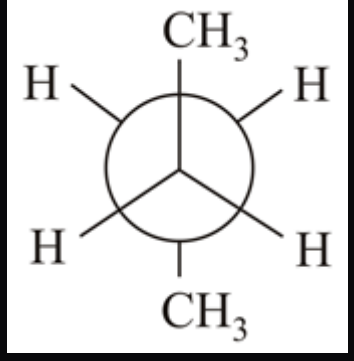


(D)



Q.38

In the given conformation C_2 is rotated about $C_2 - C_3$ bond anticlockwise by an angle of 120°, then the conformation obtained is



(A) fully eclipsed conformation.

(B) partially eclipsed conformation.

(C) gauche conformation.

(D) staggered conformation.

Q.39

How many litres of oxygen (at STP) are required for complete combustion of 39 g of liquid benzene? (Atomic weights: $\mathrm{C} = 12, \mathrm{O} = 16, \mathrm{H} = 1$)

(A) 84

(B) 22.4

(C) 42

(D) 11.2

Q.40

Which of the following process is not used for the purification of solid impurities?

(A) Distillation

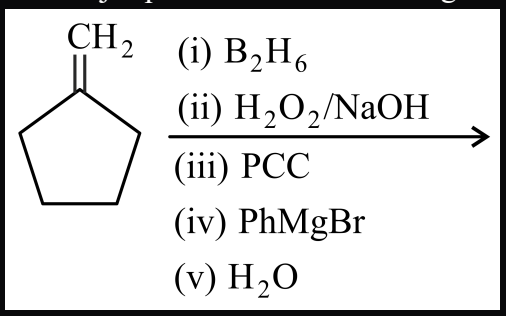
(B) Sublimation

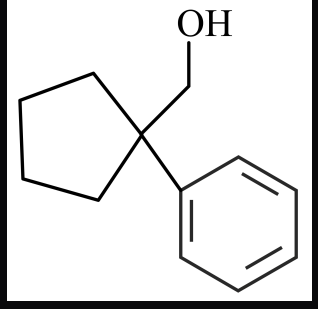
(C) Crystallisation

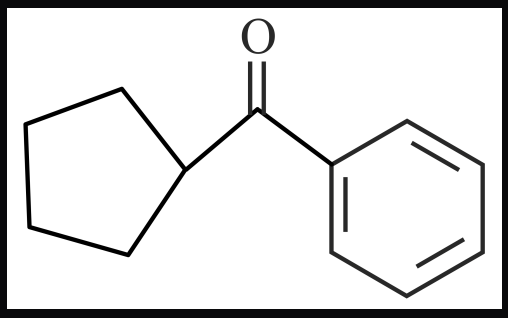
(D) Vaporisation

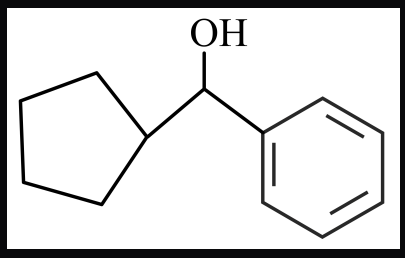
Q.41

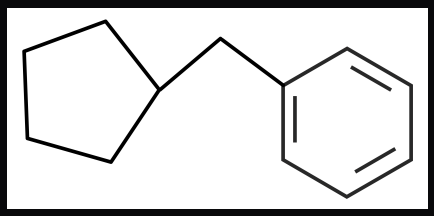
The major product of the following reactions is



(A) 

(B) 

(C) 

(D) 

Q.42

Pick out the correct statement set for transition metals

1. They form interstitial compounds
2. 5d-elements have higher energies than $3\,d$ or $4\,d$ -elements
3. Zr and Hf have almost identical atomic and ionic radii
4. They have low melting and boiling points (or low enthalpies of atomization)

(A) 1,2

(B) 2,3

(C) $1, 2, 3$

(D) $1, 2, 3, 4$

Q.43

Assertion (A): The denaturation of proteins can destroy all $1^{\circ}, 2^{\circ}$ and 3° protein structures.
Reason (R): Curdling of milk is due to denaturation of proteins.
The correct option among the following is

(A) (A) is true, (R) is true and (R) is the correct explanation for (A)

(B) (A) is true, (R) is true but (R) is not the correct explanation for (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Q.44

The structure of PCC is

(A) $C_6H_5\overset{\oplus}{N}HCrO_2Cl^{\ominus}$

(B) $C_6H_5\overset{\oplus}{N}HCrO_3Cl^{\ominus}$

(C) $C_5H_5\overset{\oplus}{N}HCrO_2Cl^{\ominus}$

(D) $C_5H_5\overset{\oplus}{N}HCrO_3Cl^{\ominus}$

Q.45

The hybridisation of metals involved in the following complexes, respectively are $[Mn(CN)_6]^{3-}$, $[CoF_6]^{3-}$

(A) sp^3d^2, sp^3d^2

(B) sp^3d^2, d^2sp^3

(C) d^2sp^3, d^2sp^3

(D) d^2sp^3, sp^3d^2

Q.46

A rigid and insulated tank of $5\ m^3$ volume is divided into the two compartments. One compartment of volume $4\ m^3$ contains an ideal gas at $0.8314\ MPa$ and $400\ K$ while the second compartment of volume $1\ m^3$ contains the same gas at $8.314\ MPa$ and $500\ K$. If the partition between the two compartments is ruptured, the final temperature of the gas will be (Mark answer to nearest whole number)

Q.47

The maximum number of atoms that can be in one plane in the molecule p-nitrobenzonitrile are

Q.48

Observe the following complexes

$[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{-3}$, $[Co\,(C_2O_4)_3]^{3-}$, $[MnCl_6]^{3-}$,

$[Fe(CN)_6]^{4-}$, $[CoF_6]^{3-}$

From the above, the number of inner orbital complexes with paramagnetic nature is

Q.49

The number of Faradays required to completely deposit magnesium from $1\ L$ of $0.1M\ MgCl_2$ aq. solution is K, find 20K

Q.50

A metal (M) forms two oxides. The ratio $\mathrm{M}:\mathrm{O}$ (by weight) in the two oxides are $25:4$ and $25:6$. The minimum value of atomic mass of M is

Q.51

$\mathrm{A}(27,-243,81)$ is a point in space. $\mathrm{B}, \mathrm{C}, \mathrm{D}$ are images of A with respect to XY, YZ and ZX planes respectively. If the centroid of the triangle BCD is (α, β, γ), then $\alpha+\beta+\gamma=$

(A) -25

(B) 45

(C) 25

(D) -45

Q.52

The number of real roots of the equation $\sin\left[2\cos^{-1}\left\{\cot\left(2\tan^{-1}x\right)\right\}\right]=0$ that are greater than or equal to one are

(A) 1

(B) 2

(C) 3

(D) 4

Q.53

If $\alpha \in \mathbb{R}-\{-1\}$ and $\mathrm{f}(\mathrm{x})=|(|\mathrm{x}|+\alpha)(|\mathrm{x}|-1)|$, then the number of points at which $f(x)$ is not differentiable, is

(A) 3 , when $\alpha<0$

(B) 5 , when $\alpha>0$

(C) 4 , when $\alpha>0$

(D) 5 , when $\alpha<0$

Q.54

The number of points $P(x, y)$ with natural numbers as coordinates that lie inside the quadrilateral formed by the lines $2x+y=2, x=0, y=0$ and $x+y=5$ is

(A) 12

(B) 10

(C) 6

(D) 4

Q.55

$\cos^4 \frac{\pi}{8} + \cos^4 \frac{2\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{4\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{6\pi}{8} + \cos^4 \frac{7\pi}{8} + \cos^4 \frac{8\pi}{8}$ is equal to

(A) 3

(B) -1

(C) 1

(D) 4

Q.56

If L_1 is a line passing through the point $\mathrm{P}(4, -3)$ and perpendicular to the line $3x - 4y + k = 0$, then the distance of P from the line $5x - 3y - 2 = 0$ measured along the line L_1 is

(A) 5

(B) $\sqrt{13}$

(C) $\sqrt{41}$

(D) 13

Q.57

If θ is the acute angle between the two lines whose direction cosines are connected by the relations $l + m + n = 0$ and $2lm + 2nl - mn = 0$, then $\cos\theta =$

(A) $\frac{1}{2}$

(B) $\frac{\sqrt{3}}{2}$

(C) $\frac{5}{6}$

(D) $\frac{3}{5}$

Q.58

If $f(x) = (2k + 1)x - 3 - ke^{-x} + 2e^{x}$ is monotonically increasing for all $x \in R$, then the least value of k is

(A) 1

(B) 0

(C) $-\frac{1}{2}$

(D) -1

Q.59

If $\alpha_1, \beta_1, \gamma_1, \delta_1$ are the roots of the equation $ax^4 + bx^3 + cx^2 + dx + e = 0$ and $\alpha_2, \beta_2, \gamma_2, \delta_2$ are the roots of the equation $ex^4 + dx^3 + cx^2 + bx + a = 0$ such that $0 < \alpha_1 < \beta_1 < \gamma_1 < \delta_1, 0 < \alpha_2 < \beta_2 < \gamma_2 < \delta_2$ $\alpha_1 - \delta_2 = 2 = \beta_1 - \gamma_2; \gamma_1 - \beta_2 = \delta_1 - \alpha_2 = 4$, then $a + b + c + d + e =$

(A) 10

(B) 12

(C) 6

(D) 8

Q.60

For the hyperbola $x^2 - y^2 - 4x + 2y + c = 0$, if the focus is $S(2 + 2\sqrt{2}, k)$ and the directrix that is adjacent to S is $x = 2 + \sqrt{2}$, then $c =$

(A) 0

(B) -1

(C) 1

(D) 2

Q.61

The solution of the differential equation $(2x - 4y + 3)\frac{dy}{dx} + (x - 2y + 1) = 0$ is
(C is an arbitrary constant)

(A) $\log[(2x - 4y) + 3] = x - 2y + C$

(B) $\log[2(2x - 4y) + 3] = 2(x - 2y) + C$

(C) $\log[2(x - 2y) + 5] = 2(x + y) + C$

(D) $\log[4(x - 2y) + 5] = 4(x + 2y) + C$

Q.62

If $\lim_{x \to 0} \frac{2a \sin x - \sin 2x}{\tan^3 x}$ exists and is equal to 1 then the value of a is

(A) 2

(B) 1

(C) 0

(D) -1

Q.63

If the point of intersection of the lines $\mathbf{r} = \hat{\mathbf{i}} - 6\hat{\mathbf{j}} + (p \sec \alpha)\hat{\mathbf{k}} + t(\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}})$ and $\mathbf{r} = 4\hat{\mathbf{j}} + \hat{\mathbf{k}} + \lambda(2\hat{\mathbf{i}} + (p \tan \alpha)\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$ is $8\hat{\mathbf{i}} + 8\hat{\mathbf{j}} + 9\hat{\mathbf{k}}$, (where $0 < \alpha < \frac{\pi}{2}$), then $p =$

(A) $\sqrt{5}$

(B) $\sqrt{3}$

(C) $\sqrt{2}$

(D) 0

Q.64

The least positive term of an arithmetic progression whose first two terms are $\frac{5}{2}$ and $\frac{23}{12}$ is

(A) 6

(B) 5

(C) $\frac{1}{6}$

(D) $\frac{37}{7}$

Q.65

Let $S_r = \{(x, y, z) | x + y + z = 11, x \geq r, y \geq r, z \geq r, x, y, z, r$ are integers $\}$ and $n(S_r)$ represents the number of elements in S_r. Then $n(S_{2)} + n(S_3) + n(S_4) =$

(A) 78

(B) 27

(C) 165

(D) 234

Q.66

If a, b and c are real numbers such that $a^2 + b^2 + c^2 - ab - bc - ac \leq 0$, then

$$\begin{vmatrix} (a-b+1)^5 & b^7 - c^7 & c^9 - a^9 \\ a^{11} - b^{11} & (b-c+2)^3 & c^{13} - a^{13} \\ a^{15} - b^{15} & b^{17} - c^{17} & (c-a+3)^1 \end{vmatrix} =$$

(A) $2abc$

(B) 0

(C) $24abc$

(D) 24

Q.67

The domain of the real valued function $f(x) = \frac{\sqrt{\log_{10}\left(\frac{x}{x-2}\right)}}{\sqrt{[x]^2 - 5[x] + 6}}$ is (Here $[\mathrm{x}]$ denotes the greatest integer function)

(A) $(-\infty, 0) \cup (2, \infty)$

(B) $(2, \infty)$

(C) $(-\infty, 2) \cup (4, \infty)$

(D) $[4, \infty)$

Q.68

If $\phi(t)=\begin{cases} 1, \text{ for } 0 \leq t<1 \\ 0, \text{ otherwise} \end{cases}$, then $\int_{-3000}^{3000}\left(\sum_{r'=2014}^{2016} \phi(t-r') \phi(t-2016)\right) dt$ is

(A) a real number

(B) 1

(C) 0

(D) does not exist

Q.69

If $f\left(\frac{2x+3}{3x+5}\right)=x+4, x \neq \frac{-5}{3}, \frac{2}{3}$ and $\int f(x)dx = Ax + B\ln|3x-2| + C$, then $3B-A=$

(A) $\frac{64}{9}$

(B) $\frac{-52}{21}$

(C) $\frac{-10}{3}$

(D) $\frac{-8}{3}$

Q.70

Four persons A, B, C and D throw an unbiased die, turn by turn in succession till one gets an even number and win the game. What is the probability that A wins the game if A begins ?

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{7}{15}$

(D) $\frac{8}{15}$

Q.71

If $\int_0^{\pi} \frac{\sin x(\sin x+1)e^{\sin x+\cos x}}{e^{\cos x}+1}dx = a + b\int_0^{\pi} e^{\sin x}dx$ where a and b are positive rational numbers, then find the value of $100\left(a^2+b^2\right)$.

Q.72

The standard deviations of two sets of observations $X=\{x_i\}$ and $Y=\{y_i\}$ $(i=1,2,\ldots,100)$ are respectively 5 and 6 . If $\bar{x}, \bar{y}$ are their means and $\sum_{i=1}^{100}\left(x_i-\bar{x}\right)\left(y_i-\bar{y}\right)=600$, then the standard deviation of $Z=\{z_i | z_i=x_i-y_i)$ is

Q.73

If O, G, S are respectively the orthocentre, centroid and circumcentre of a triangle whose vertices are $A(2,3), B(2,4)$ and $C(4,3)$, then $AO^2+9BG^2+4CS^2=$

Q.74

If the numerically greatest term in the expansion of $(2-3\mathrm{x})^{9}$ when $\mathrm{x}=1$ is $\mathrm{P}_{1}^{\alpha}\mathrm{P}_{2}^{\beta}\mathrm{P}_{3}^{\gamma}\mathrm{P}_{4}^{\delta}$ ($\mathrm{P}_1<\mathrm{P}_2<\mathrm{P}_3<\mathrm{P}_4$ are the first four prime numbers), then $\alpha+\beta+\gamma+\delta=$

Q.75

If the image of the point $(1,-2,1)$ with respect to the line passing through the points $\mathrm{B}(1,1,2)$ and $\mathrm{C}(2,2,1)$ is (ℓ,m,n), then $\ell^2+m^2+n^2=$

Answers & Solutions

Q.1 Answer:

128

Solution:

$$pV^{\gamma}=\frac{p}{\rho^{\gamma}}=\text{ constant}$$

$$\therefore \quad \frac{p'}{p}=\left(\frac{\rho'}{\rho}\right)=(32)^{7/5}$$

$$=32\times 4=128$$

Q.2 Answer:

0.052 cm

Solution:

Diameter of wire,

$d=\mathrm{MSR}+\mathrm{CSR}\ \times \mathrm{LC}$

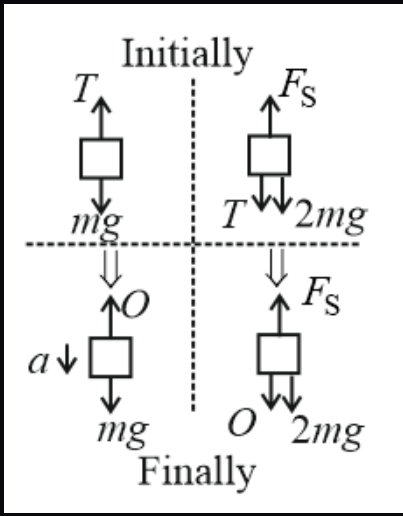
$=0+52\times\frac{1}{100}$

$=0.52\ \mathrm{mm}=0.052\ \mathrm{cm}.$

Q.3 Answer:

$\frac{g}{2}$ upwards, g downwards

Solution:



Initially, both the blocks were under equilibrium, so for mass m,

$T = mg$,

and for mass $2m$,

$F_s = T + 2mg = 3mg$

Now, after the string is cut, the tension in it becomes zero. Therefore,

for m,

$mg - 0 = ma \Rightarrow a = g$**(downward)**

and for $2m$,

$2mg + 0 - F_s = 2mb2mg + 0 - 3mg = 2mb \Rightarrow b = -\frac{g}{2}$

i.e., $\frac{g}{2}$ upward.

Q.4 Answer:

$31.85 \times 10^{-5}\ \mathrm{m}$

Solution:

Young modulus, $Y = 2 \times 10^{11}\ \mathrm{N\ m^{-2}}$

When boy jumps on free end of wire, tension produced in wire, T =weight of boy,

$T = mg = 10 \times 10 = 100\ \mathrm{N}$

Here, m is the mass of boy.

Stress= $\frac{F}{A} = \frac{F}{\pi r^2} = \frac{F}{\pi\left(\frac{d}{2}\right)^2}$

where $F = T$, is force produced in wire, A is area of cross section, r is the radius of wire and d is the diameter of wire.

Stress $= \frac{100}{3.14 \times \left(\frac{2 \times 10^{-3}}{2}\right)^2} = \frac{100 \times 10^6}{3.14}$

As Young's modulus,

$Y = \frac{\text{stress}}{\text{strain}} = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta l}{l}\right)}$

where Δl is change in length, l is length of wire.

$\Delta l = \frac{\text{stress} \times l}{Y} = \frac{100 \times 10^6 \times 2}{3.14 \times 2 \times 10^{11}}$

$= 31\ .85\ \times 10^{-5}\ \mathrm{m}$

Q.5 Answer:

1.0×10^{-17} kg ms^{-1}

Solution:

$\text{Power} = \frac{\text{E}}{\text{t}}$ $\text{E} = \text{Energy}$

$\therefore$ $\text{E} = \text{Power} \times \text{time} = \frac{\text{hc}}{\lambda}$

$\frac{\text{hc}}{\lambda} = 30 \times 10^{-3} \times 100 \times 10^{-9}$

$\therefore$ $\text{p} = \frac{\text{h}}{\lambda} = \frac{3 \times 10^{-9}}{3 \times 10^{8}} = 1 \times 10^{-17}$ kg ms

Q.6 Answer:

$t^{3/2}$

Solution:

Power, $P = Fv =$ **constant**

Or $ma(at) =$ **constant**

Or $a^2 t =$ **constant**

$\Rightarrow a \propto \frac{1}{\sqrt{t}} \ldots\ldots\ldots(1)$

Now, $S = \sqrt{2}$ **at** $^2 (\because u = 0)$

$\Rightarrow S \propto at^2$

$\therefore S \propto \frac{t^2}{\sqrt{t}} [\text{Using}(1)]$

$\Rightarrow S \propto t^{3/2}$

Q.7 Answer:

(A) is not correct but (R) is correct.

Solution:

Key Idea In nature, metals have positive temperature coefficient and semiconductors have negative temperature coefficient.

Aluminium is a metal, so its conductivity decreases with increase in temperature, due to the excessive collision of electron. Whereas, Si **is a semiconductor therefore its conductivity increases with increase in temperature.**

All metals have positive temperature coefficient of resistivity and semiconductor has negative temperature coefficient of resistivity. Hence, assertion (A) is incorrect but reason (R) is correct.

Q.8 Answer:

$2s$

Solution:

$F_{\text{rest}} = \left(\frac{GmM}{R^3}\right) r$

$\Rightarrow$ **From equation (i)** $; T = 2\pi \sqrt{\frac{R^3}{GM}}$

$\Rightarrow T = 2\pi \sqrt{\frac{R^3}{10R^3}}$

$\Rightarrow T = 2 \sec$

Q.9 Answer:

0.3

Solution:

Initial speed of body on the horizontal rough surfaces, $u = 10\ \text{ms}^{-1}$

Final velocity, $v = 4\ \text{ms}^{-1}$ and $t = 2\ \text{s}$.

If a be the retardation due to friction, then by using equation,

$v = u - at$

$\Rightarrow \quad 4 = 10 - a \times 2$

$\Rightarrow \quad 2a = 6 \Rightarrow a = 3\ \text{ms}^{-2}$

According to given situation,

friction force $= ma$

$\Rightarrow \quad \mu_k mg = ma$

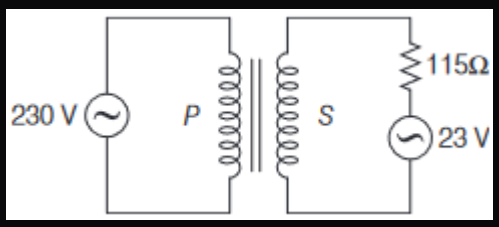
$\Rightarrow \quad \mu_k = \dfrac{a}{g} = \dfrac{3}{10} = 0.3$

Q.10 Answer:

0.02 A

Solution:

The given situation is shown in the figure



For the given transformer,

$V_p = 230\ \text{V}$

$V_S = 23\ \text{V}$

$R_S = 115\Omega$

Currrent in secondary coil,

$I_S = \frac{V_S}{R_S} = \frac{23}{115} = 0.2\ \text{A}$

We know that, in a transformer

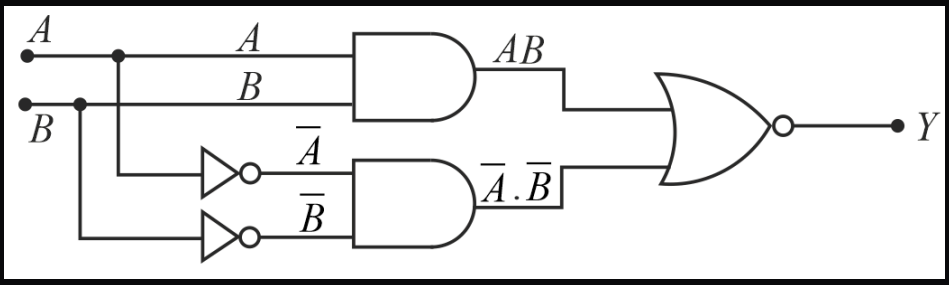
$$\frac{V_S}{V_P} = \frac{I_P}{I_S}$$

$$\Rightarrow \quad I_P = \frac{V_S I_S}{V_P} = \frac{23 \times 0.2}{230} = 0.02\ \text{A}$$

Q.11 Answer:

0, 0

Solution:



In the circuit, the output wil be given by,

$Y = \overline{A \cdot B + \overline{A} \cdot \overline{B}}$

When both inputs are 1means $A = B = 1$, then the output will be,

$Y = \overline{1 \cdot 1 + \overline{1} \cdot \overline{1}}$

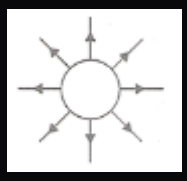
$Y = \overline{1} = 0$.

Similarly, for both input are 0, the output will be,

$Y = 0$

Hence, the output will be 0, 0.

Q.12 Answer:



Solution:

For a uniformly charged hollow cylinder, the electric field lines are as shown in figure (b).

Q.13 Answer:

$\frac{3}{32} I_0$

Solution:

Since, the pass axes of two polarisers P_1 and P_2 were kept such that the incident unpolarised beam of intensity I_0 gets completely blocked.

This means that both polarisers were kept cross (perpendicular of their pass axes) to each other. When third polariser P_3 is introduced between P_1 and P_2 such that the angle between the pass axes of P_1 and P_3 is $60°$, i.e. $\theta_1 = 60°$.

intensity of polarised light emerging from first polaroid, $I_1 = \frac{I_0}{2}$

Intensity of polarised light (I_3) emerging from polaroid P_3 is given by law of Malus, i.e.,

$$I_3 = I_1 \cos^2 \theta,$$

$$= \frac{I_0}{2} \cos^2 60 = \frac{I_0}{2} \frac{1}{4}$$

$$I_3 = \frac{I_0}{8}$$

Now, the angle between pass axis of P_3 and P_2, $\theta_2 = 90° - 60° = 30°$

$\therefore$ Intensity of polarised light emerging from lost (second) polaroid P_2.

$I_2 = I_3 \cos^2 30° = \frac{I_0}{8} \cdot \frac{3}{4} = \frac{3I_0}{32}$

Q.14 Answer:

A, B true C false

Solution:

Time period of satellite moving around a planet with angular velocity ω is given as,

$T = \frac{2\pi}{\omega}$

Since, the two artificial satellites revolving in the same circular orbit, then their angular velocities will be same.

Hence, both satellites revolve with same period of revolution.

Orbital velocity of revolving satellite, $v = \sqrt{\frac{Gm}{r}}$

i.e $v \propto \frac{1}{\sqrt{r}}$

Where, $m =$ **mass of the planet**

$G =$ **Universal gravitational constant**

$r =$ **radius of circular orbit.**

Escape velocity depends on the gravitational potential at the point from where the body is launched. Since, this potential depends on the height of the point of projection, hence escape velocity also depends on the height (altitude) of the point of projection.

Q.15 Answer:

Assertion is true statement but Reason is false.

Solution:

In terms of mass-energy interconversion, a chemical reaction is conceptually equivalent to a nuclear reaction. The difference in chemical (not nuclear) binding energies of atoms and molecules on the two sides of a reaction can be traced to the energy emitted or consumed in a chemical reaction.

A chemical reaction's mass defects, on the other hand, are about a million times smaller than those in a nuclear reaction.

Q.16 Answer:

$\frac{1}{6}\sqrt{\frac{\rho_2}{\rho_1}}$

Solution:

Frequency of closed organ pipe for first harmonic $n_1 = \frac{v_1}{4l_1}$.

Frequency of open organ pipe for third harmonic

$n_3 = \frac{3v_2}{2l_2}$

At resonance, $\quad n_1 = n_3$

or $\quad \frac{v_1}{4l_1} = \frac{3v_2}{2l_2}$

or $\quad \frac{l_1}{l_2} = \frac{1}{6}\left(\frac{v_1}{v_2}\right)$

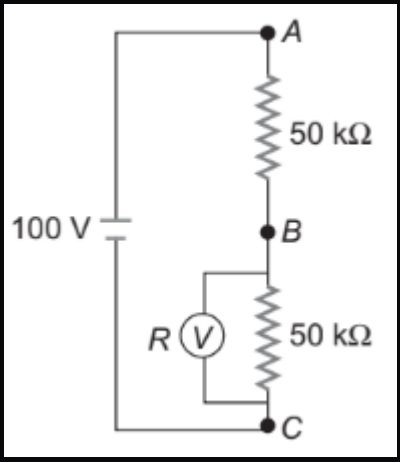
$\frac{l_1}{l_2} = \frac{1}{6}\sqrt{\frac{B}{\rho_1}} \times \sqrt{\frac{\rho_2}{B}}$

$= \frac{1}{6}\sqrt{\frac{\rho_2}{\rho_1}}$

Q.17 Answer:

$50\text{k}\Omega$

Solution:



Internal resistance of voltmeter is R.

Therefore effective resistance across B and C, R' is given by

$$\frac{1}{R'} = \frac{1}{R} + \frac{1}{50}$$

$$= \frac{50 + R}{50R}$$

or $R' = \left(\frac{50R}{50+R}\right)$

According to Ohm's law

$$V' = IR'$$

$$\text{or} \quad \frac{100}{3} = I \cdot \left(\frac{50R}{50 + R}\right)$$

$$\text{or} \quad \frac{100}{3}\left(\frac{50 + R}{50R}\right) = I \qquad \ldots(\text{i})$$

Now, total resistance of circuit

$$R'' = 50 + \frac{50R}{50 + R}$$

$$\text{or} \quad R'' = \frac{(2500 + 100R)}{(50 + R)}$$

$$\text{Now,} \quad V'' = IR''$$

$$\Rightarrow \quad 100 = \tfrac{100}{3}\left(\tfrac{50+R}{50R}\right)\tfrac{2500+100R}{(50+R)}$$

$$\text{or} \quad 150R = 2500 + 100R$$

$$\text{or} \quad 50R = 2500$$

$$\text{or} \quad R = 50\text{k}\Omega$$

Q.18 Answer:

$E_\gamma > E_\text{x} > E_\text{v}$

Solution:

$E_\gamma > E_\mathrm{x} > E_\mathrm{v}$

Gamma rays, a form of nuclear and cosmic EM radiation, can have the highest frequencies and, hence, the highest photon energies in the EM spectrum.The range of photon energies for visible light from red to violet is 1.63 to$3.26\ \mathrm{eV}$ .An X-ray, or X-radiation, is a penetrating form of high-energy electromagnetic radiation. Most X-rays have a wavelength ranging from$10^{-12}\ \mathrm{m}$ to $10^{-9}\ \mathrm{m}$, corresponding to frequencies in the range $30\ \mathrm{petahertz}$ to $30\ \mathrm{exahertz}\ (30 \times 10^{15}\ \mathrm{Hz}\ \mathrm{to}\ 30 \times 10^{18}\ \mathrm{Hz})$ $124\ \mathrm{eV}\ \mathrm{to}\ 124\ \mathrm{keV}$and energies in the range .

Q.19 Answer:

$\mathrm{ML}^4\,\mathrm{T}^{-2}$

Solution:

Given, $PV^{\frac{5}{3}} = K$

Pressure $P\ = \left[\mathrm{ML}^{-1}\,\mathrm{T}^{-2}\right]$

Volume $V = \left[L^3\right]$

$\Rightarrow \left[\mathrm{ML}^{-1}\,\mathrm{T}^{-2}\right]\left[L^3\right]^{5/3} = K$

$\therefore$ **Dimensions of** $K = \mathrm{ML}^4\,\mathrm{T}^{-2}$

Q.20 Answer:

A and R and true and R is correct explanation of A.

Solution:

When an ideal gas is compressed adiabatically, its temperature and the average kinetic energy of the gas molecule increases because of collision of molecules with wall. Hence, Both A and R are true and R is the correct explanation of A.

Q.21 Answer:

—

Solution:

In $L - C - R$ **circuit,**

$X_L = 2R$ and $X_C = \frac{X_L}{3}$

Impedance of $L - C - R$ **circuit is given as**

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{R^2 + \left(X_L - \frac{X_L}{3}\right)^2}$$

$$= \sqrt{R^2 + \left(\frac{2X_L}{3}\right)^2} = \sqrt{R^2 + \frac{4X_L^2}{9}}$$

$$= \sqrt{R^2 + \frac{4}{9}(2R)^2} = \sqrt{R^2 + \frac{16}{9}R^2}$$

$$= \sqrt{\frac{25R^2}{9}} \Rightarrow Z = \frac{5}{3}R$$

$$\text{Power factor } = \cos\phi$$

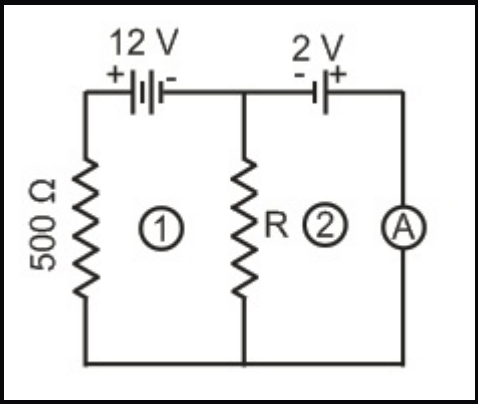
$$= \frac{R}{Z} = \frac{R}{5R/3} = \frac{3}{5} = 0.6$$

Q.22 Answer:

—

Solution:

Given circuit



In loop (1)

$$12 - 500i_1 - Ri_1 = 0$$

$\Rightarrow 12 = i_1(500 + R)$ **...(i)**

In loop (2)

$$12 - 500i_1 - 2 = 0$$

$\Rightarrow 10 = 500i_1$

or $i = \frac{1}{50}\text{A}$ **...(ii)**

From eqs. (i) and (ii),

$12 \times \frac{1}{i_1} = (500 + \text{R})$

$\Rightarrow 12 \times 50 = 500 + \text{R}$

$\Rightarrow \text{R} = 100\Omega$

Q.23 Answer:

—

Solution:

Given, energy of hydrogen atom in ground state $= -|E|$ and $n_A = 3$

$\therefore$ Energy of electrons in excited state, n_A

$(E)_{n_A} = \frac{-|E|}{n_A^2}$

Energy of electrons in excited state, n_B

$= (E)_{nB} = \frac{-(E)}{n_B^2}$

When sample absorbs the photon of energy, $\frac{|E|}{12}$, then its electrons reaches from energy state n_A to energy state n_B.

$$\text{Hence, } (E)_{n_B} - (E)_{n_A} = \frac{|E|}{12}$$

$$\frac{-|E|}{n_B^2} - \left\{\frac{-|E|}{n_A^2}\right\} = \frac{|E|}{12}$$

$$\frac{-1}{n_B^2} + \frac{1}{n_A^2} = \frac{1}{12} \Rightarrow -\frac{1}{n_B^2} = \frac{1}{12} - \frac{1}{n_A^2}$$

$$= \frac{1}{12} - \frac{1}{9} \Rightarrow -\frac{1}{n_B^2} = -\frac{1}{36}$$

$$\Rightarrow \quad n_B^2 = 36$$

$$\therefore \quad n_B = 6$$

Q.24 Answer:

—

Solution:

$$m = 0.6\ \text{g} = 0.6 \times 10^{-3}\ \text{kg}, q = 25\text{nC} = 25 \times 10^{-9}\text{C}$$

$$v = 1.2 \times 10^4\ \text{m/s}$$

The particle is moving with uniform velocity (acceleration is zero), so magnetic force will be balancing the weight of the particle.

$$mg = Bqv$$

$$0.6 \times 10^{-3} \times 10 = B \times 25 \times 10^{-9} \times 1.2 \times 10^4$$

$$B = \frac{0.6 \times 10^{-3} \times 10}{25 \times 10^{-9} \times 1.2 \times 10^4}$$

$$= \frac{6 \times 10^{-3}}{30 \times 10^{-5}} = \frac{6 \times 10^2}{30}$$

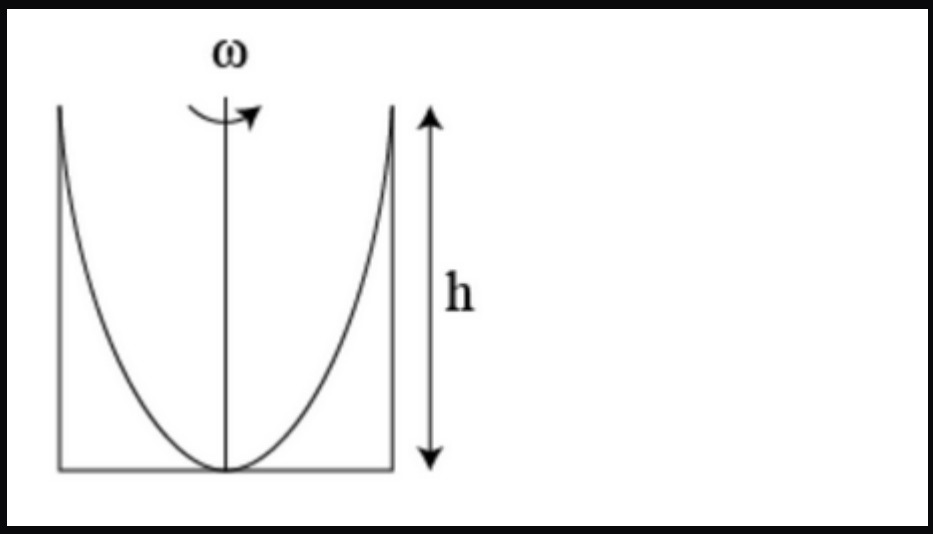
$$= 20\ \text{T}$$

Q.25 Answer:

—

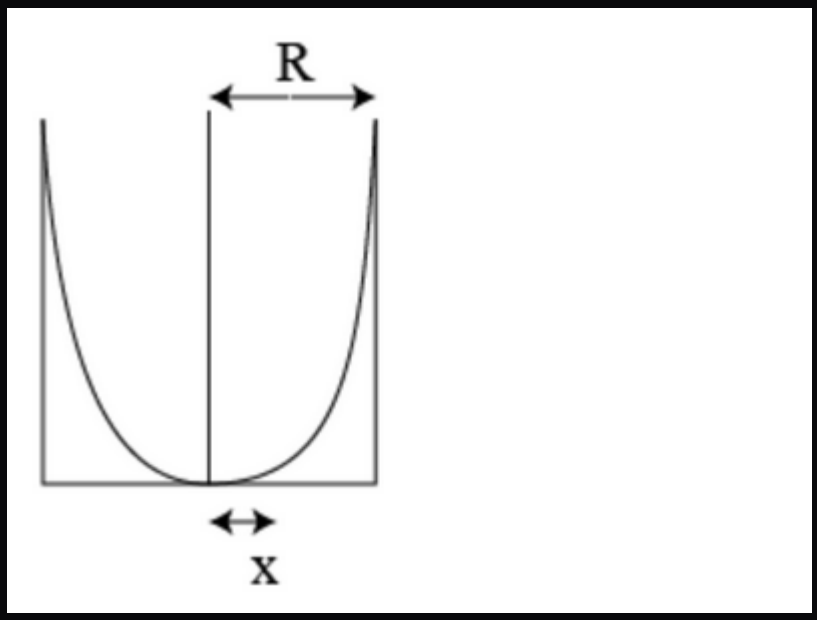
Solution:

Let ω_1 be initial angular velocity



$\frac{\omega_1^2 R^2}{2g} = h$

For angular velocity ω_2 let radius of exposed surface be x



$\frac{\omega_2^2}{2g}\left[R^2 - x^2\right] = h$

Putting the value of ω_1 and $\omega_2 \pi x^2 = 9$

Since $\pi R^2 = 27$

Q.26 Answer:

(v) (i) (ii) (iii)

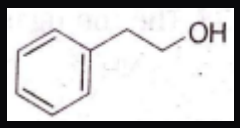
Solution:

EDTA is used in the determination of hardness of water.

Chlorophyll is used in photosynthesis. Oxyhaemoglobin contains Fe^{2+} ion, so it is paramagnetic and haemoglobin works as oxygen carrier.

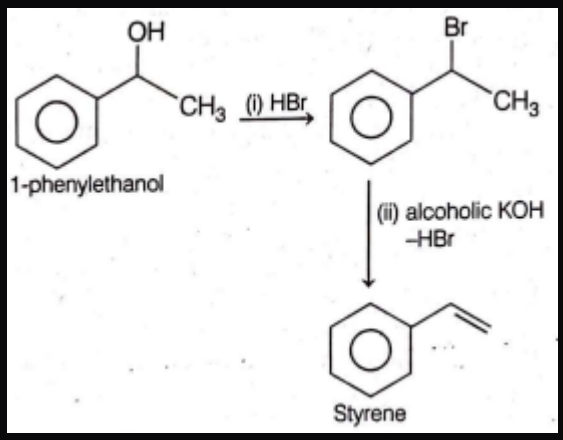
Hence, A-(v), B-(i), C-(ii), D-(iii).

Q.27 Answer:

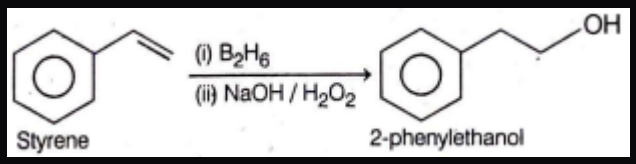


Solution:

Firstly, 1-phenylethanol undergoes nucleophillic substitution reaction with HBr to give alkyl halide which further undergoes elimination reaction in presence of alcohols KOH to give styrene.



Styrene further undergoes hydroboration oxidation reaction to give 2-phenylethanol.



Q.28 Answer:

H^+

Solution:

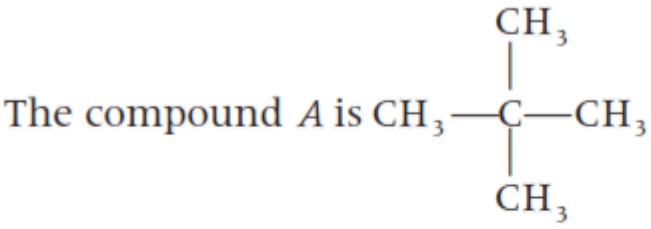
The exposure of iron (or an alloy of iron) to oxygen in the presence of moisture leads to the formation of rust. This reaction is not instantaneous, it generally proceeds over a considerably large time frame. The rusting of iron occurs in acidic medium. Hence, it is catalyzed by H^+ ions.The oxygen atoms bond with iron atoms, resulting in the formation of iron oxides. This weakens the bonds between the iron atoms in the object/structure.

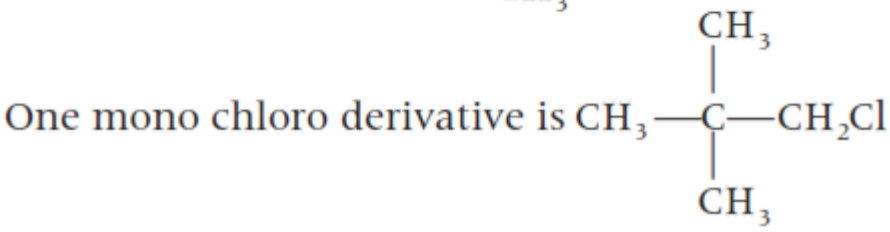
Q.29 Answer:

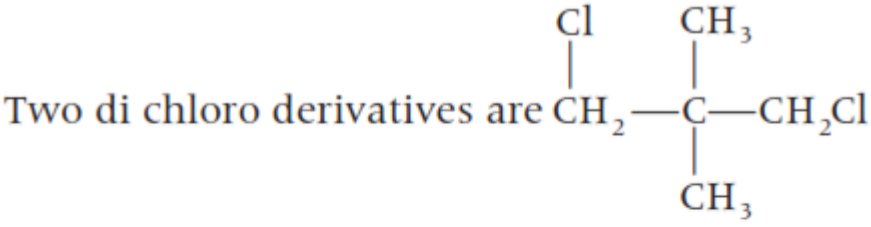
2, 2-dimethylpropane

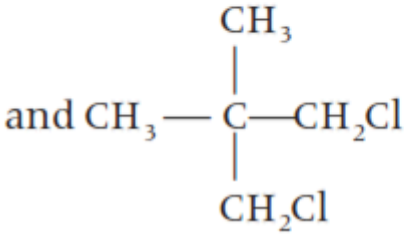
Solution:

C_5H_{12}, pentane has molecular mass 72 g i.e. the isomer of pentane which yields single monochloro derivative and two dichloro derivative is 2, 2-dimethyl propane as it has all equivalent hydrogens.

The compound *A* is 

One mono chloro derivative is 

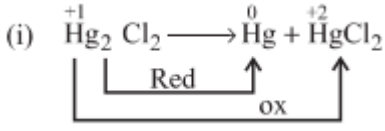
Two di chloro derivatives are 

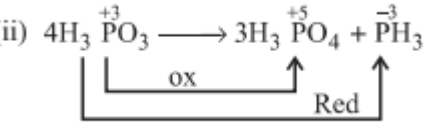
and 

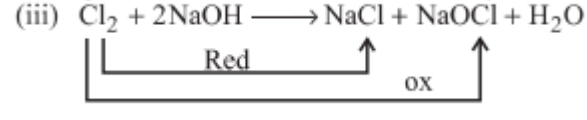
Q.30 Answer:

$BaO_2 + H_2SP_4 \longrightarrow BaSO_4 + H_2O_2$

Solution:

(i) 

(ii) 

(iii) 

(iv) $BaO_2 + H_2SO_4 \rightarrow BaSO_4 + H_2O_2$; **Displacement reaction.**

Q.31 Answer:

$H_2O > H_2Te > H_2Se > H_2S$

Solution:

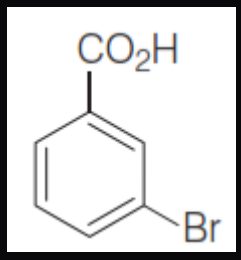
All the elements of Group 16 form hydrides of the type H_2E (E = O, S, Se, Te, Po). Their acidic character increases from H_2O to H_2Te.

Boiling point depends on the size and intermolecular forces.

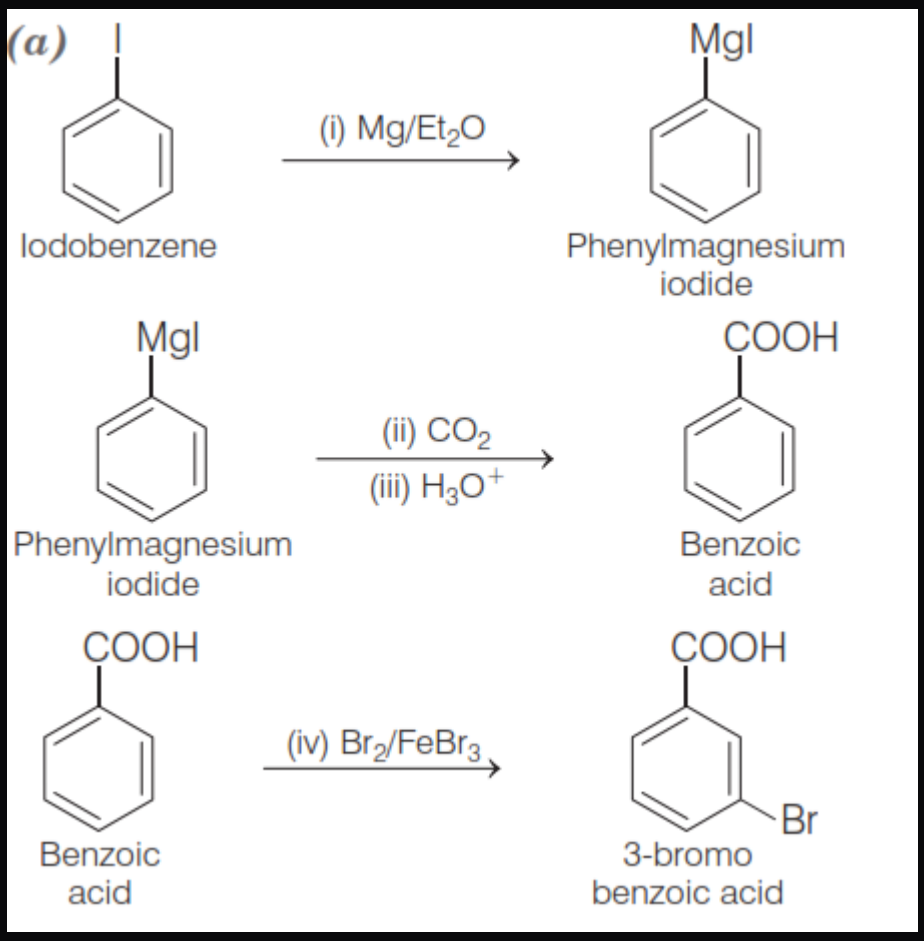
As the size of the central atom increases boiling point should increase,but H_2O has greater boiling point than the other hydrides due to intermolecular hydrogen bonding.

The correct order of boiling points of 16th group hydrides is $H_2O > H_2Te > H_2Se > H_2S$The most volatile hydride of 16th group is H_2S

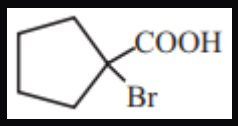
Q.32 Answer:



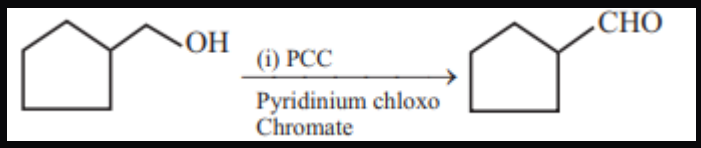
Solution:



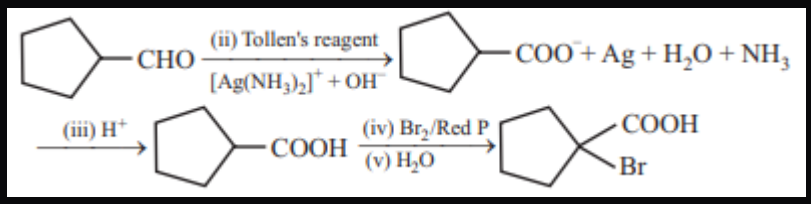
Q.33 Answer:



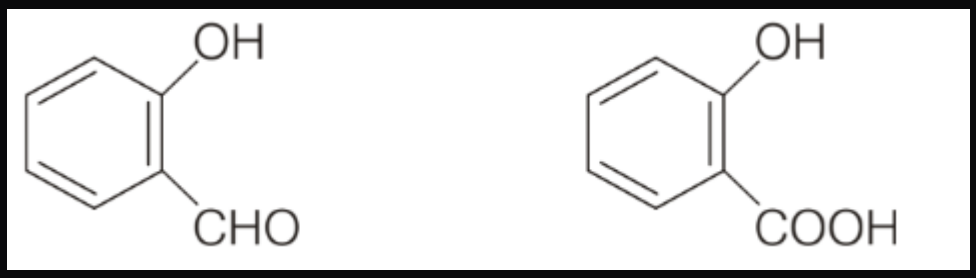
Solution:



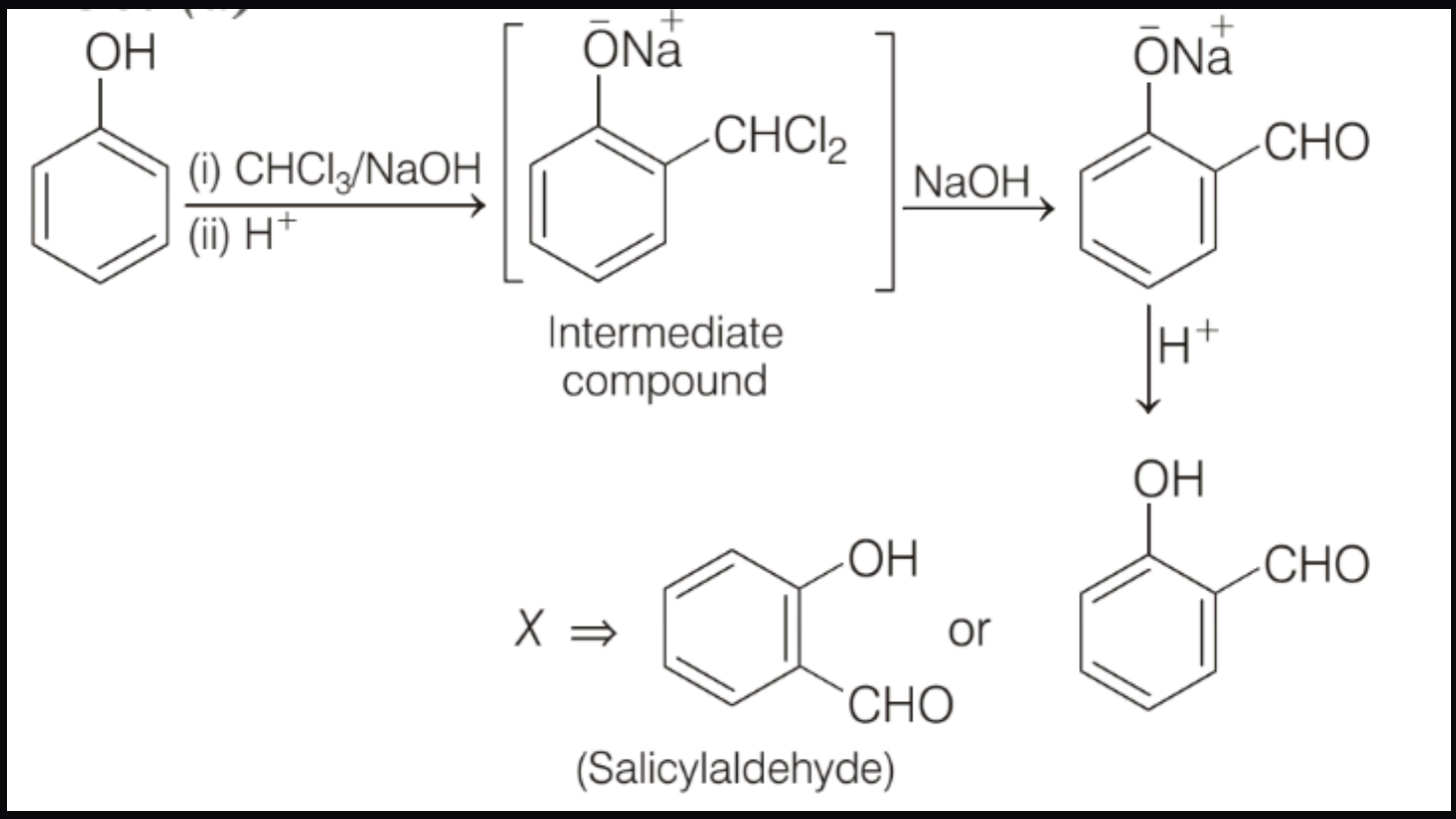
Best method for preparing aldehyde from primary alcohol.



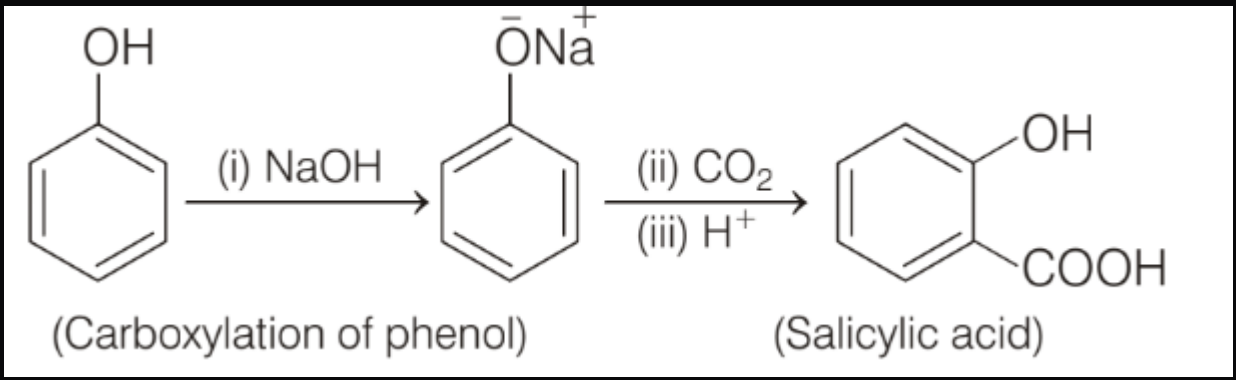
Q.34 Answer:



Solution:



When phenol treated with chloroform $(CHCl_3)$ in the presence of aqueous sodium hydroxide $(NaOH)$, product will be 2-hydroxy benzaldehyde (salicylaldehyde).



When phenol react with $CO_2, NaOH$, it gives salicylic acid (o-hydroxy benzoic acid).

Q.35 Answer:

$n - C_6H_{14} + n - C_7H_{16}$

Solution:

An ideal solution obeys Raoult's law at all temperature and pressure.
The solute-solute and solvent-solvent interaction are almost similar to solute-solvent interaction. No association or dissociation takes place. Water and methanol does not form azeotropic mixture and shows position deviation. Chloroform and benzene show negative deviations. Only n-hexane and n-heptane mixture form an ideal solution.

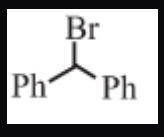
Q.36 Answer:

Assertion is correct, Reason is wrong.

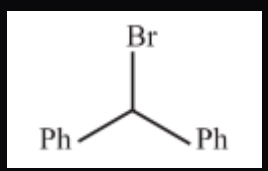
Solution:

Actinides exhibit larger oxidation states than lanthanides because of the very small energy gap between $5f, 6d$ and $7s$ subshells. Thus, the outermost electrons get easily excited to the higher energy levels, giving variable oxidation states. Hence, Assertion is correct, Reason is incorrect. Thus, the correct option is (3).

Q.37 Answer:



Solution:



Forms the carbocation is S_N1 mechanism, which gets the stability by delocalization of positive charge through two benzene rings.

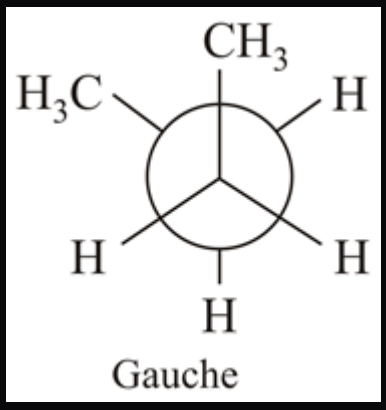
Q.38 Answer:

gauche conformation.

Solution:

For forming conformers we have to see torsional angle with respect to a group. Here we assume front carbon on one side and rotates back carbon so when we reach to angle 60° & 300° we get gauche conformation.

Now with given question we get:



This structure is at torsional angle of 300°.

Q.39 Answer:

84

Solution:

$$C_6H_6 + 7\tfrac{1}{2}O_2 \longrightarrow 6CO_2 + 3H_2O$$
$$78\ g \qquad 7.5 \times 22.4\ L$$

Volume of O_2 required for 78 g of C_6H_6

$= 7.5 \times 22.4\ L$

Volume of O_2 required for $39\ g of C_6H_4$

$= \frac{7.5 \times 22.4 \times 39}{78} = 84L.$

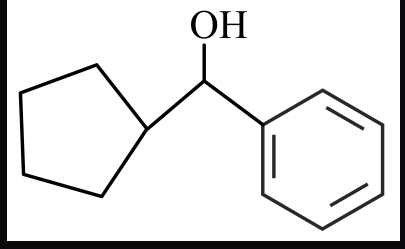
Q.40 Answer:

Distillation

Solution:

Distillation process is not used for purification of solid impurities. It is used for the purification of liquids which boils without decomposition and contains non-volatile impurities.

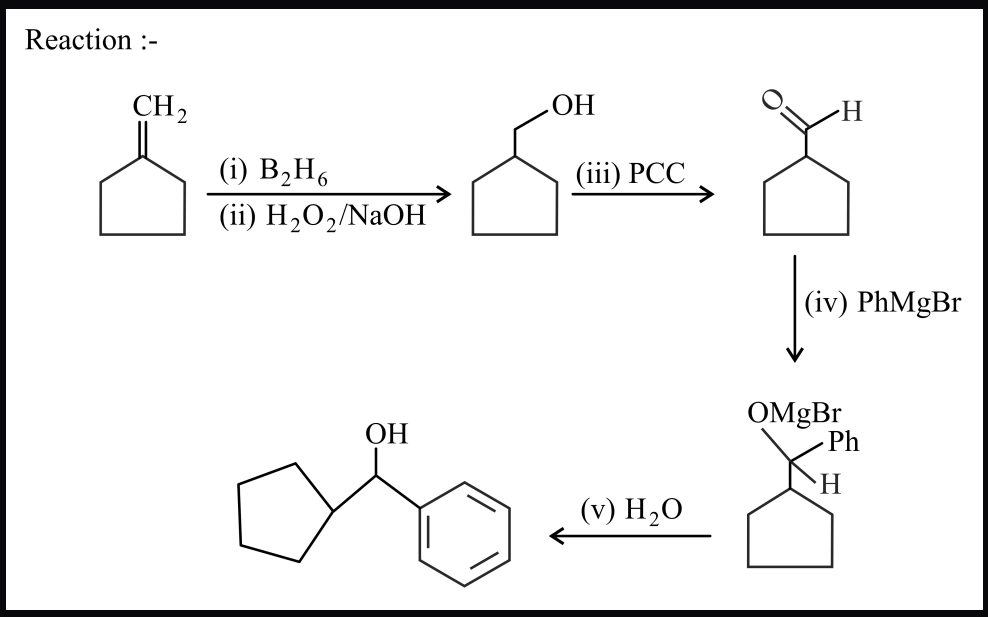
Q.41 Answer:



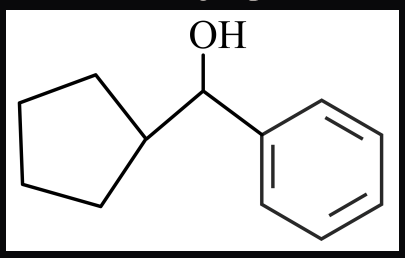
Solution:

Alkene gives the hydroboration oxidation reaction. It formed the alkene to alcohol. After that the alcohol is oxidised into the aldehyde by the PCC. After that Grignard reagent followed by the hydrolysis gives the nucleophilic addition reaction. It converts the aldehyde into the secondary alcohol.

Reaction :-



Hence, the major product of this reaction is



.

Q.42 Answer:

$1, 2, 3$

Solution:

Here statement 1,2 and 3 are correct as follows

Because small size, atoms of H, C, N occupy the vacant spaces in the crystal lattice of transition metal.

This due to the greater effective nuclear charge acting on outer valence electrons, because of weak shielding of nucleus by 4 f-electrons

The normal size increase from Zr to Hf is almost exactly balanced by lanthanide contraction

Transition metals have high enthalpies of atomization, because they form strong metallic bonds, due to participation of $(n-1)$ d-electrons along with ns-electrons in metallic bonding.

Q.43 Answer:

(A) is false but (R) is true

Solution:

Protein found in a biological system with a unique three-dimensional structure and biological activity is called a native protein. When a protein in its native form, is subjected to physical change like change in temperature or chemical change like change in pH, the hydrogen bonds are disturbed. Due to this, globules unfold and helix get uncoiled and protein loses its biological activity. This is called denaturation of protein.

The coagulation of egg white on boiling is a common example of denaturation. Another example is curdling of milk which is caused due to the formation of lactic acid by the bacteria present in milk. During denaturation secondary and tertiary structures are destroyed but primary structure remains intact.

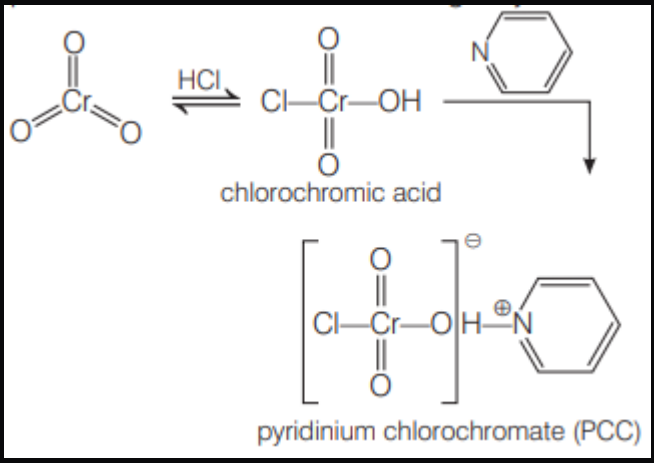
Hence,(A) is false but (R) is true

Q.44 Answer:

$C_5H_5\overset{\oplus}{N}HCrO_3Cl^{\ominus}$

Solution:

Pyridinium chlorochromate (PCC) can be prepared by the dissolution of chromium trioxide in aqueous HCl. Addition of pyridine gives pridinium chlorochromate as orange cryotol.



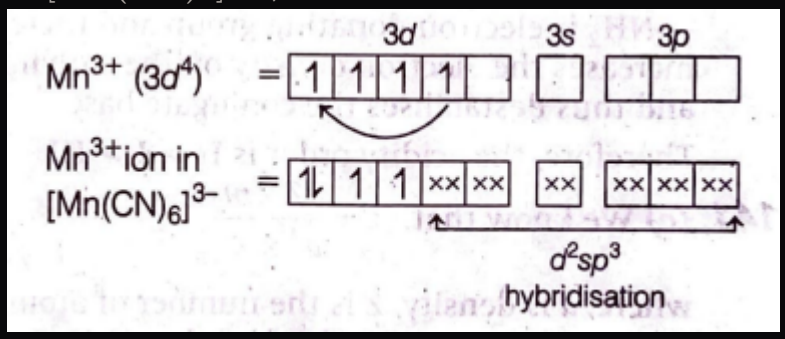
PCC offers the advantage of the selective oxidation of alcohols to aldehydes, whereas many other reagents are less selective.

Q.45 Answer:

d^2sp^3, sp^3d^2

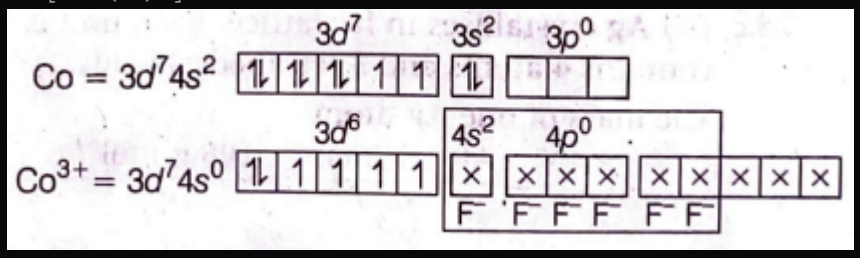
Solution:

In $[Mn(CN)_6]^{3-}$, Mn is in +3 oxidation state.



As CN^- is a strong field ligand, it causes pairing of two $3d$ electrons, thereby making two d-orbitals vacant, namely $d_{x^2-y^2}$ and d_{z^2}. These two d-orbitals and one $4s$ and three $4p$-orbitals hybridise and give d^2sp^3-hybridisation.

In $[Co(F)_6]^{3-}$, Co is in +3 oxidation state.

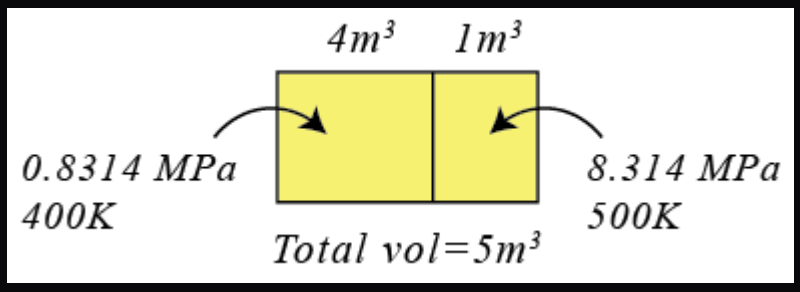


As F^- is a weak field ligand, so it cannot cause the pairing of unpaired $3d$-electrons. Thus, one $4s$, three $4p$-orbitals and two $4d$-orbitals hybridise and give sp^3d^2-hybridisation.

Q.46 Answer:

—

Solution:



$$n_1 = \frac{P_1V_1}{RT_1} = \frac{0.8314 \times 10^6 \times 4}{8.314 \times 400} = 1000$$

$$n_2 = \frac{P_2V_2}{RT_2} = \frac{8.314 \times 10^6 \cdot 1}{8.314 \times 500} = 2000$$

$\Delta V = 0$

$w = -P_{\text{ext}}\,\Delta V = 0$

$q = 0$

$\Delta U = 0$

$\Delta U = nC_V\Delta T$

$1000C_v\,(T_f - 400) + 2000C_v\,(T_f - 500) = 0$

$T_f - 400 + 2T_f - 1000 = 0$

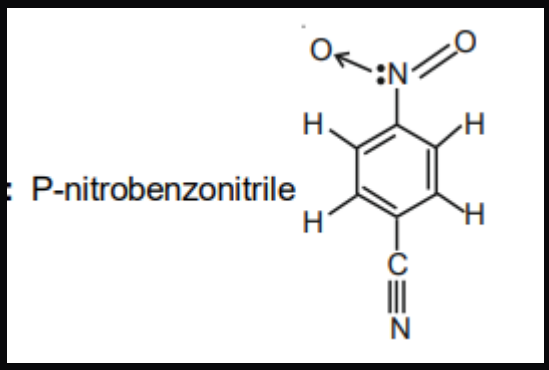
$3T_f = 1400$

$T_f = 466.6 \approx 467$

Q.47 Answer:

—

Solution:



All atoms are on the same plane as all central atoms are either sp^2 or sp hybridised $\therefore$ total number of atoms on same plane $\Rightarrow \mathrm{C_6H_4} + \mathrm{NO_2} + \mathrm{CN} = 15$

Q.48 Answer:

—

Solution:

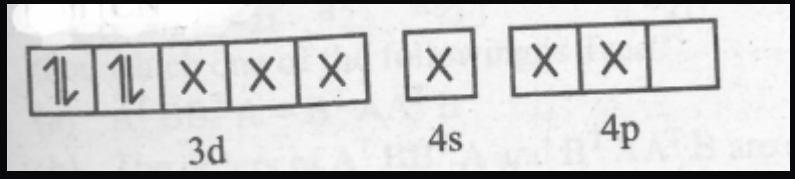
$[\mathrm{Mn(CN)_6}]^{3-}: -$

$\mathrm{Mn}^{3+} = [\mathrm{Ar}]^3\ \mathrm{d}^4$

CN^- is a strong-field ligand so it pairs the electrons.

Thus,

$[\mathrm{Mn(CN)_6}]^{3-} =$



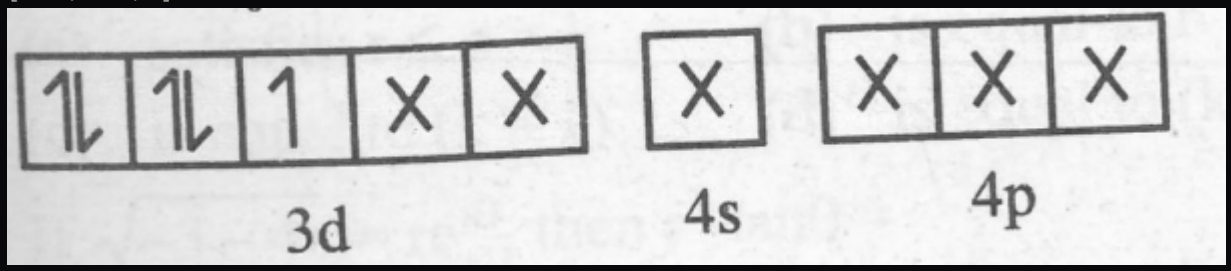
Thus, it is an inner-orbital but a diamagnetic complex.

$[Fe(CN)_6]^{3-}$: –

$Fe^{3+} = [Ar]3\,d^5$

CN^- **will pair - up the electrons so :-**

$[Fe(CN)_6]^{3-} =$

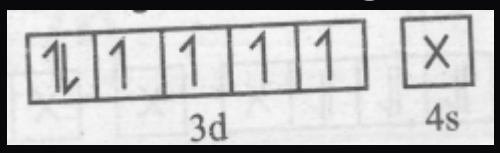


Thus, it is an inner-orbital and a paramagnetic complex.

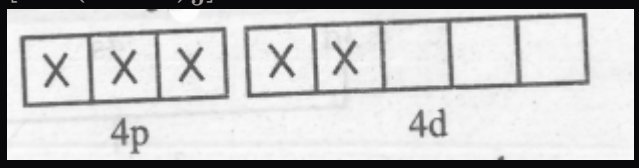
$[Co\,(C_2O_4)_3]^{3-}$: –

$Co^{3+} = [Ar]3\,d^6$

$C_2O_4{}^{2-}$ **is a weak-field ligand so :-**



$[Co\,(C_2O_4)_3]^{3-} =$

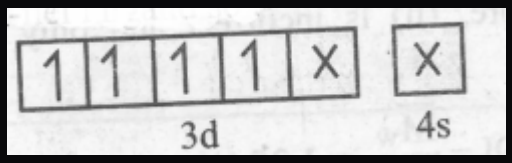


Thus, it is an outer-orbital and paramagnetic complex

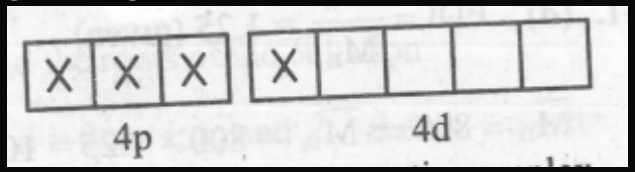
$[MnCl_6]^{3-}$: –

$Mn^{3+} = [Ar]3\,d^4$

Cl^- **is a weak-field ligand so :-**



$[MnCl_6]^{3-} =$

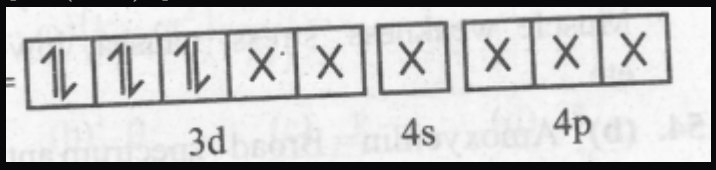


Thus, it is an inner-orbital and paramagnetic complex $[Fe(CN)_6]^4$ **:-**

$Fe^{2+} = [Ar]3\,d^6$ **and** CN^- **is a strong-field ligand.**

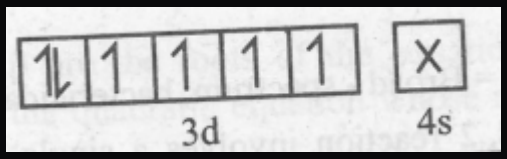
Thus,

$[Fe(CN)_6]^{4-} =$

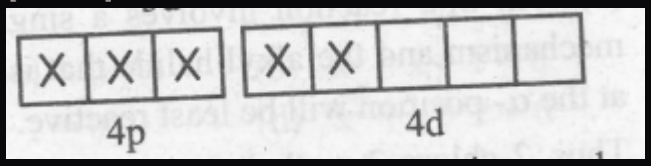


Thus, it is an inner-orbital and diamagnetic complex. $[CoF_6]^{3-} : -\ Co^{3+} = [Ar]3\ d^6$ **and** F^- **is a weak-field ligand.**

Thus,



$[CoF_6]^{3-} =$



Thus, it is an outer-orbital and paramagnetic complex. Therefore, there are two inner-orbital and paramagnetic complexes.

Q.49 Answer:

—

Solution:

$MgCl_2 \rightarrow Mg^{2+} + 2Cl^-$

$Mg^{2+} + 2e^- \rightarrow Mg\ (s)$

2 moles of e^- **are required to deposit 1 mole of** $Mg(s)$**.**

$\Rightarrow 1$ mole of $e^- = 1$ F

$\Rightarrow 2$ moles of $e^- = 2$ F

$n\ (MgCl_2) = 0.1M \times 1\ L = 0.1$ mol.

$n\left(Mg^{2+}\right) = 0.1$ mol.

Therefore, the number of moles of electrons/faradays required to deposit 0.1 mol **of** Mg **will be** $2 \times 0.1 = \mathbf{0.2mol}$**.**

Q.50 Answer:

—

Solution:

Let two oxide be

M_2O_x **and** M_2Oy

As per question

$\frac{2a}{16x} = \frac{25}{4}$ and $\frac{2a}{16y} = \frac{25}{6}$

$x = \frac{a}{50}, y = \frac{3a}{100}$ where $a =$ atomic mass of Metal

As x and y to be an integer,

If we take $a = 50$, then $x = 1, y = 1.5$ (not possible)

If we take $a = 100$ then $x = 2, y = 3$ (possible)

$\therefore$ Minimum Atomic Mass $= 100\text{u}$

Q.51 Answer:

-45

Solution:

Given point $\text{A}(27, -243, 81)$ in space, now images of point A in xyh plane, yzh plane and zx-planes are B $(27, -243, -81), C(-27, -243, 81)$ and $D(27, 243, 81)$ respectively.

Hence centroid of the triangle BCD is $= (9, -81, 27) \equiv (\alpha, \beta, \gamma)$ hence, $\alpha + \beta + \gamma = -45$

Q.52 Answer:

2

Solution:

Given equation

$\sin\left[2\cos^{-1}\left\{\cot\left(2\tan^{-1}x\right)\right\}\right] = 0$

$\Rightarrow 2\cos^{-1}\left\{\cot\left(2\tan^{-1}x\right)\right\} = n\pi, n \in \mathbf{Z}$

$\Rightarrow \cos^{-1}\left\{\cot\left(2\tan^{-1}x\right)\right\} = \frac{n\pi}{2}$

$\because \quad \cos^{-1}x \in [0, \pi]$

$\therefore \quad \cos^{-1}\left\{\cot\left(2\tan^{-1}x\right)\right\} = 0, \frac{\pi}{2}, \pi$

$\Rightarrow \cot\left(2\tan^{-1}x\right) = 1, 0, -1 \Rightarrow 2\tan^{-1}x = m\pi + \frac{\pi}{4}$

$m\pi + \frac{\pi}{2}$ or $m\pi - \frac{\pi}{4}, m \in \mathbf{Z}$

$\Rightarrow \tan^{-1}x = \frac{m\pi}{2} \pm \frac{\pi}{8}$ or $\frac{m\pi}{2} + \frac{\pi}{4}, m \in \mathbf{Z}$

$\because \quad \tan^{-1}x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\therefore \tan^{-1}x = \pm\frac{\pi}{8}, \frac{\pi}{2} - \frac{\pi}{8}, -\frac{\pi}{2} + \frac{\pi}{8}, \pm\frac{\pi}{4}$

$\therefore \quad x = \pm\tan\frac{\pi}{8}, \pm\cot\frac{\pi}{8}, \pm\tan\frac{\pi}{4}$

$\Rightarrow \quad x = \pm(\sqrt{2} - 1), \pm\frac{1}{\sqrt{2}-1}, \pm 1$

$= \pm(\sqrt{2} - 1), \pm(\sqrt{2} + 1), \pm 1$

The roots which are greater than or equal to one are $\sqrt{2} + 1, 1$.

Q.53 Answer:

5 , when $\alpha < 0$

Solution:

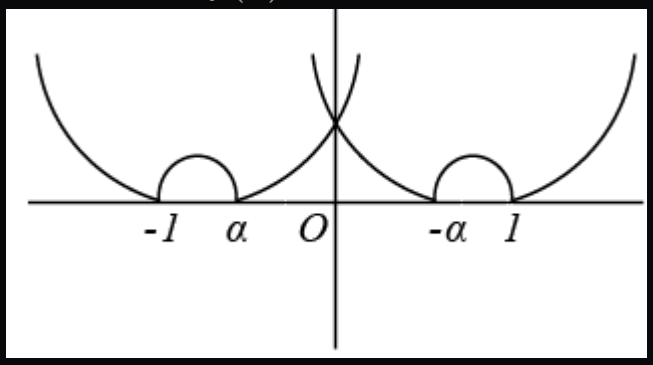
Since, $\alpha \in R - \{-1\}$

and $\text{f(x)} = |(|x| + \alpha)(|x| - 1)|$

$|x| + \alpha = 0 \Rightarrow |x| = -\alpha \Rightarrow \alpha < 0$

$|x| - 1 = 0 \Rightarrow x = \pm 1$

Now, graph of $f(x)$ is



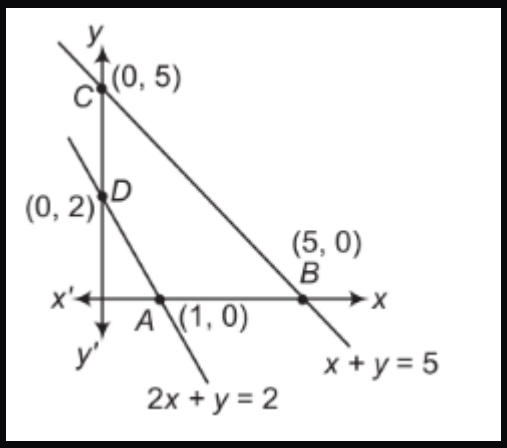
It is clear from graph that there are 5 points at which $\mathrm{f}(\mathrm{x})$ is not differentiable and $\alpha < 0$

Q.54 Answer:

6

Solution:

Given equations are $2x + y = 2, x = 0, y = 0$ and $x + y = 5$.



The total integral points lies inside the quadrilateral $ABCD$ is

$(1, 1), (1, 2), (2, 1), (2, 2), (1, 3), (3, 1)$

Q.55 Answer:

3

Solution:

$$\cos^4\frac{\pi}{8}+\cos^4\frac{2\pi}{8}+\cos^4\frac{3\pi}{8}+\cos^4\frac{4\pi}{8}$$
$$+\cos^4\frac{5\pi}{8}+\cos^4\frac{6\pi}{8}+\cos^4\frac{7\pi}{8}+\cos^4\frac{8\pi}{8}$$
$$=\left(\cos^4\frac{\pi}{8}+\cos\frac{7\pi}{8}\right)+\left(\cos^4\frac{2\pi}{8}+\cos^4\frac{6\pi}{8}\right)$$
$$+\left(\cos^4\frac{3\pi}{8}+\cos^4\frac{5\pi}{8}\right)+\left(\cos^4\frac{4\pi}{8}+\cos^4\frac{8\pi}{8}\right)$$
$$=\left(2\cos^4\frac{\pi}{8}+2\cos^4\frac{2\pi}{8}+2\cos^4\frac{3\pi}{8}\right)+\cos^4\frac{\pi}{2}+\cos^4(\pi)$$
$$=2\left[\cos^4\frac{\pi}{8}+\cos^4\left(\frac{\pi}{2}-\frac{\pi}{8}\right)+\cos^4\frac{\pi}{4}\right]+0+1$$
$$=2\left(\cos^4\frac{\pi}{8}+\sin^4\frac{\pi}{8}+\frac{1}{4}\right)+1$$
$$=2\left\{\left(\cos^2\frac{\pi}{8}+\sin^2\frac{\pi}{8}\right)^2-2\sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}+\frac{1}{4}\right\}+1$$
$$=2\left\{1-2\sin^2\frac{\pi}{8}\cos^2\frac{\pi}{8}+\frac{1}{4}\right\}+1$$
$$=2-\left(2\sin\tfrac{\pi}{8}\cos\tfrac{\pi}{8}\right)^2+\frac{1}{2}+1=3-\left(\frac{1}{\sqrt{2}}\right)^2+\frac{1}{2}$$
$$=3$$

Q.56 Answer:

5

Solution:

Slope of line perpendicular to $3x-4y+k=0$ **is** $\frac{-4}{3}$

$\therefore L_1: y+3=\frac{-4}{3}(x-4)\Rightarrow 3y+9=-4x+16$

$L_1: 4x+3y=7$ **and** $L: 5x-3y=2$ **(given)**

Point of intersection of L_1 **and** $L: (x,y)\equiv(1,1)$

$\therefore$ **Distance between** $P(4,-3)$ **and** $(1,1)$

$=\sqrt{(4-1)^2+(-3-1)^2}=5.$

Q.57 Answer:

$\frac{1}{2}$

Solution:

$l+m+n=0$
$2lm+2nl-mn=0$
$l=-(m+n)$
$\Rightarrow -2m(m+n)-2n(m+n)-mn=0$
$\Rightarrow -2m^2-2mn-2mn-2n^2-mn=0$
$\Rightarrow -2m^2-2n^2-5mn=0$
$\Rightarrow 2m^2+5mn+2n^2=0$
$\Rightarrow \quad (2m+n)(m+2n)=0$
$\Rightarrow n=-2m, m=-2n$
when $n=-2m$
$l=-(m+n)=-(m-2m)=m$
$\therefore \quad l=m=\dfrac{-n}{2}$
$\Rightarrow \quad \dfrac{l}{1}=\dfrac{m}{1}=\dfrac{n}{-2}$
$\therefore \quad$ **D.R.** $\equiv (1,1,-2)$
when $n=-2m$
$l=-(m+n)=-(m-2m)=m$
$\therefore \quad l=m=\dfrac{-n}{2}$
$\Rightarrow \quad \dfrac{l}{1}=\dfrac{m}{1}=\dfrac{n}{-2}$
$\therefore \quad$ D.R. $\equiv (1,1,-2)$
when $m=-2n$
$l=-(m+n)=-(-2n+n)=n$
$\therefore \quad l=n=\dfrac{m}{-2}$
$\Rightarrow \quad \dfrac{l}{1}=\dfrac{m}{-2}=\dfrac{n}{1}$
$\therefore \quad$ D.R. $\equiv (1,-2,1)$
$\therefore \quad \cos\theta=\left|\dfrac{1-2-2}{\sqrt{1+1+4}\cdot\sqrt{1+4+1}}\right|=\dfrac{3}{6}=\dfrac{1}{2}.$

Q.58 Answer:

0

Solution:

Given,
$f(x)=(2k+1)x-3-ke^{-x}+2e^{x}$
Since, $f(x)$ **is monotonically increasing for all** $x\in R$.
So,

$$
\begin{aligned}
& & (2k+1)+ke^{-x}+2e^{x} & \geq 0 \\
\Rightarrow & \quad e^{-x}\left((2k+1)e^{x}+k+2e^{2x}\right) & \geq 0 & \\
\text{or} & \quad (2k+1)e^{x}+k+2e^{2x} & \geq 0 & \\
\Rightarrow & \quad 2e^{x}\left(e^{x}+k\right)+1\left(e^{x}+k\right) & \geq 0 & \\
\Rightarrow & & \left(2e^{x}+1\right)\left(e^{x}+k\right) & \geq 0 \\
\Rightarrow & & e^{x}+k \geq 0 \text{ or } k & \geq 0
\end{aligned}
$$

Hence, least value of k is zero.

Q.59 Answer:

8

Solution:

Given,

$\alpha_1, \beta_1, \gamma_1, \delta_1$ are the roots of equation

$ax^4+bx^3+cx^2+dx+e=0$ and

$\alpha_2, \beta_2, \gamma_2, \delta_2$ are the roots of equation

$ex^4+dx^3+cx^2+bx+a=0$

$\therefore$ Clearly,

$\alpha_1, \beta_1, \gamma_1, \delta_1$ are the reciprocal of roots of

$ax^4+bx^3+cx^2+dx+e=0$

Also, $\delta_1>\gamma_1>\beta_1>\alpha_1>0$

and $\delta_2>\gamma_2>\beta_2>\alpha_2>0$

$\therefore$ δ_1 and α_2 are reciprocal

Similarly γ_1 and β_2,

γ_2 and β_1 and α_1 and δ_2 are reciprocal

$$
\begin{aligned}
\therefore \quad & \alpha_1\delta_2=1, \beta_1\gamma_2=1, \beta_2\gamma_1=1, \alpha_2\delta_1=1 \\
& \alpha_1-\delta_2=2 \\
\therefore \quad & \alpha_1-\tfrac{1}{\alpha_1}=2 \Rightarrow \alpha_1^2-2\alpha_1-1=0 \\
& \text{and} \quad \delta_1-\alpha_2=4 \\
& \delta_1-\tfrac{1}{\delta_1}=4=\delta_1^2-4\delta_1-1=0
\end{aligned}
$$

$\therefore$ α_1 and δ_1 are roots of

$$
\begin{aligned}
& ax^4+bx^3+cx^2+dx+e=0 \\
\therefore & \left(x^2-2x-1\right)\left(x^2-4x-1\right)=ax^4+bx^3+cx^2+dx+e
\end{aligned}
$$

Put $x=1$,

$$
\begin{aligned}
(1-2-1)(1-4-1) &= a+b+c+d+c \\
8 &= a+b+c+d+e
\end{aligned}
$$

Q.60 Answer:

-1

Solution:

Given hyperbola

$$x^2 - y^2 - 4x + 2y + c = 0$$

$$(x-2)^2 - (y-1)^2 = 3 - c$$

$$\frac{(x-2)^2}{3-c} - \frac{(y-1)^2}{3-c} = 1$$

Focus of hyperbola is

$$x - 2 = \pm\sqrt{(3-c)+(3-c)}$$

$$x - 2 = \pm\sqrt{6-2c} \quad [\because ae = \sqrt{(3-c)+(3-c)}]$$

$$x = 2 \pm \sqrt{6-2c}$$

Here, $2 + \sqrt{6-2c} = 2 + 2\sqrt{2}$

$$\therefore \quad \sqrt{6-2c} = 2\sqrt{2} \Rightarrow 6 - 2c = 8 \Rightarrow c = -1$$

Q.61 Answer:

$\log[4(x-2y)+5] = 4(x+2y) + C$

Solution:

Given differential equation,

$$(2x - 4y + 3)\frac{dy}{dx} + (x - 2y + 1) = 0$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{(x-2y+1)}{2(x-2y+3)} \quad \ldots \text{(i)}$$

Let, $x - 2y = v \Rightarrow 1 - 2\frac{dy}{dx} = \frac{dy}{dx}$

$\therefore \frac{1}{2}\left(1 - \frac{dy}{dx}\right) = \frac{dy}{dx}$

Now, substitute $v = x - 2y$ **and** $\frac{dy}{dx}$ **in eq. (i) integration both the sides, we get**

$$\therefore \quad \frac{1}{2}\left(1 - \frac{dv}{dx}\right) = -\left(\frac{v+1}{2v+3}\right)$$

$$\Rightarrow \quad \frac{dv}{dx} = 1 + \frac{2v+2}{2v+3}$$

$$\Rightarrow \frac{2v+2}{4v+5}dv = dx$$

$$\frac{1}{2}\int \frac{4v+6}{4v+5}dv = \int dx$$

$$\Rightarrow \frac{1}{2}\int \left(\frac{4v+6}{4v+5} + \frac{1}{4v+5}\right)dv = \int dx$$

$$\Rightarrow \frac{1}{2}v + \frac{1}{2\times 4}\log[4v+5] = x + C$$

$$\Rightarrow 4v + \log[4v+5] = 8x + C$$

$$\Rightarrow 4(x-2y) + \log[4(x-2y)+5] = 8x + C$$

$$\Rightarrow \log[4(x-2y)+5] = 4(x+2y) + C$$

Q.62 Answer:

1

Solution:

$\lim_{x\to 0}\ \frac{2a\sin x-\sin 2x}{\tan^3 x}$

$= \lim_{x\to 0}\ \frac{2a\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-\ldots\right)-\left(2x-\frac{(2x)^3}{3!}+\frac{(2x)^5}{5!}-\ldots\right)}{\left(x+\frac{x^3}{3}+\frac{2}{15}x^5+\ldots\right)^3}$

$= \lim_{x\to 0}\ \frac{\left(2a-2x+\left(-\frac{2a}{3!}+\frac{2^3}{3!}\right)x^3+\left(\frac{2a}{5!}-\frac{2^5}{5!}\right)x^5+\ldots\right.}{x^3\left(1+\frac{x^2}{3}+\frac{2}{15}x^4+\ldots\right)^3}$

since, it is given that given limit is exist, then

$2a-2=0 \Rightarrow \alpha=1$

Q.63 Answer:

$\sqrt{3}$

Solution:

It is given that the lines

$\mathbf{r}=\hat{\mathbf{i}}-6\hat{\mathbf{j}}(p\sec\alpha)\hat{\mathbf{k}}+t(\hat{\mathbf{i}}+2\hat{\mathbf{j}}+\hat{\mathbf{k}})$

and $\mathbf{r}=4\hat{\mathbf{j}}+\hat{\mathbf{k}}+\lambda(2\hat{\mathbf{i}}+(p\tan\alpha)\hat{\mathbf{j}}+2\hat{\mathbf{k}})$

Intersected at point $8\hat{\mathbf{i}}+8\hat{\mathbf{j}}+9\hat{\mathbf{k}}$**, so** $1+t=8, -6+2t=8$ **and** $p\sec\alpha+t=9$

$\therefore \quad t=7$ **and** $p\sec\alpha=2\ldots(\mathrm{i})$

and $2\lambda=8, 4+\lambda p\tan\alpha=8$ **and** $1+2\lambda=9$

$\therefore \quad \lambda=4$ **and** $p\tan\alpha=1\ldots(\mathrm{ii})$

From Eqs. (i) and (ii), we get

$p^2\sec^2\alpha-p^2\tan^2\alpha=4-1 \Rightarrow p^2=3 \Rightarrow p=\pm\sqrt{3}$

Q.64 Answer:

$\frac{1}{6}$

Solution:

$T_n=\frac{5}{2}+(n-1)\left(\frac{23}{12}-\frac{5}{2}\right)$

$=\frac{5}{2}+(n-1)\frac{(-7)}{12}=\frac{30-7n+7}{12}$

$T_n>0 \Rightarrow \frac{37-7n}{12}>0$

$\Rightarrow n<\frac{37}{7} \Rightarrow n=5$

$T_5=\frac{37-35}{12}=\frac{1}{6}$

Q.65 Answer:

27

Solution:

Given,

$S_r = \{(x, y, z) : x + y + z = 11, x \geq r, y \geq r, z \geq r\}$

$S_2 = \{(x, y, z) : x + y + z = 11, x \geq 2, x \geq 2, z \geq 2\}$

$\because x \geq 2 \Rightarrow x - 2 \geq 0, y - 2 \geq 0, z - 2 \geq 0$

Let $x - 2 = a, y - 2 = b, z - 2 = c$

$\therefore x + y + z = 11 \Rightarrow a + b + c = 5, a, b, c \geq 0$

$n(S_2) = {}^{5+3-1}C_{3-1} = {}^{7}C_2 = 21$

Similary, for $S_3, a + b + c = 2$

$n(S_3) = {}^{2+3-1}C_2 = {}^{4}C_2 = 6$

For $S_4, a + b + c = -1$ (not possible)

$\therefore \quad n(S_2) + n(S_3) + n(S_4) = 21 + 6 + 0 = 27$

Q.66 Answer:

24

Solution:

Given, $a^2 + b^2 + c^2 - ab - bc - ca \leq 0$

$\therefore \frac{1}{2}\left[(a - b)^2 + (b - c)^2 + (c - a)^2\right] \leq 0$

It is possible only $a = b = c$

$$\text{Now, } \begin{vmatrix} (a - b + 1)^5 & b^7 - c^7 & c^9 - a^9 \\ a^{11} - b^{11} & (b - c + 2)^3 & c^{13} - a^{13} \\ a^{15} - b^{15} & b^{17} - c^{17} & (c - a + 3)1 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 3 \end{vmatrix} = 24$$

Q.67 Answer:

$[4, \infty)$

Solution:

$f(x) = \frac{\sqrt{\log_{10} \frac{x}{x-2}}}{\sqrt{[x]^2 - 5[x] + 6}}$ (given)

$[x]^2 - 5[x] + 6 > 0$

$\Rightarrow ([x] - 3)([x] - 2) > 0$

$\Rightarrow [x] < 2 \text{ or } [x] > 3 \Rightarrow x \in (-\infty, 2) \cup [4, \infty)$

$\text{and } \log_{10} \frac{x}{x-2} \geq 0 \Rightarrow \frac{x}{x-2} \geq 1 \Rightarrow \frac{x}{x-2} - 1$

$\Rightarrow \frac{2}{x-2} \geq 0 \quad \therefore x \in (2, \infty)$

Taking common solution $\therefore x \in [4, \infty)$.

Q.68 Answer:

1

Solution:

$\int_{-3000}^{3000}\left(\sum_{r'=2014}^{2016}\phi\left(t-r'\right)\phi(t-2016)\right)dt$
$=\int_{-3000}^{3000}[\phi(t-2014)\phi(t-2016)+\phi(t-2015)\phi(t-2016)+\phi(t-2016)\phi(t-2016)]\mathrm{dt}$
$=\int_{-3000}^{3000}\phi(t-2016)[\phi(t-2014)+\phi(t-2015)+\phi(t-2016)]dt$
$=\int_{-3000}^{2016}0dt+\int_{2016}^{2017}1\cdot(0+0+1)dt+\int_{2017}^{3000}0dt$
$=0+1+0=1$ **which is a real number.**

Q.69 Answer:

$\frac{-8}{3}$

Solution:

We have, $f\left(\frac{2x+3}{3x+5}\right)=x+4$

$\text{Put, }\frac{2x+3}{3x+5}=y\Rightarrow 2x+3=3xy+5y\Rightarrow x=\frac{5y-3}{2-3y}$

$\therefore f(y)=\frac{5y-3}{2-3y}+4=\frac{5y-3+8-12y}{2-3y}\Rightarrow f(y)=\frac{7y-5}{3x-2}$

$\therefore\quad f(x)=\frac{7x-5}{3x-2}\Rightarrow\int f(x)dx=\int\frac{7x-5}{3x-2}dx$

Put, $3x-2=t\Rightarrow x=\frac{t+2}{3}$

$dx=\frac{dt}{3}=\frac{1}{3}\int\frac{7\left(\frac{t+2}{3}\right)-5}{t}dt$

$=\frac{1}{9}\int\frac{7t-1}{t}dt=\frac{7}{9}\int dt-\frac{1}{3}\int\frac{dt}{t}$

$=\frac{7}{9}t-\frac{1}{9}\log t+C$

$=\frac{7}{9}(3x-2)-\frac{1}{9}\log|3x-2|+C$

$=\frac{7}{3}x-\frac{1}{9}\log|3x-2|+C$

Here, $A=\frac{7}{3},B=-\frac{1}{9}$

$\therefore\quad 3B-A=3\left(-\frac{1}{9}\right)-\frac{7}{3}=-\frac{8}{3}$

Q.70 Answer:

$\frac{8}{15}$

Solution:

$\mathrm{P}(\text{ even })=\frac{1}{2},\mathrm{P}(\text{ odd })=\frac{1}{2}$
$\mathrm{P}(\text{ A wins })=\frac{1}{2}+\left(\frac{1}{2}\right)^4\times\frac{1}{2}+\left(\frac{1}{2}\right)^8\times\frac{1}{2}+\left(\frac{1}{2}\right)^{12}\times\frac{1}{2}+\ldots\ldots\ldots\ldots\ldots\ldots\infty$
$=\frac{\frac{1}{2}}{1-\left(\frac{1}{2}\right)^4}=\frac{1}{2}\times\frac{16}{15}=\frac{8}{15}$

Q.71 Answer:

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Solution:

Using King

$$I=\int_0^{\pi}\frac{\sin x(1+\sin x)e^{\sin x-\cos x}}{e^{-\cos x}+1}dx$$

$$I=\int_0^{\pi}\frac{\sin x(1+\sin x)e^{\sin x}}{e^{\cos x}+1}dx \quad \ldots(1)$$

$$\text{And } I=\int_0^{\pi}\frac{\sin x(1+\sin x)e^{\sin x}\left(e^{\cos x}+1\right)}{e^{\cos x}+1}dx$$

$$I=\frac{1}{2}\int_0^{\pi}\sin x(1+\sin x)e^{\sin x}dx$$

$$I=\frac{1}{2}\int_0^{\pi}e^{\sin x}\left(\sin x+1-\cos^2 x\right)dx$$

$$I=\frac{1}{2}\int_0^{\pi}e^{\sin x}dx+\frac{1}{2}\int_0^{\pi}e^{\sin x}\left(\sin x-\cos^2 x\right)dx$$

$$I=\frac{1}{2}\int_0^{\pi}e^{\sin x}dx+\frac{1}{2}[1+1]$$

$$I=1+\frac{1}{2}\int_0^{\pi}e^{\sin x}(-\cos x)\Big]_0^{\pi}$$

$$\therefore \quad 100\left(1+\tfrac{1}{4}\right)=125$$

Q.72 Answer:

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Solution:

From the given information, we can say that

$\sqrt{\frac{\sum_{i=1}^{100}(x_i-\bar{x})^2}{100}}=5$ and $\sqrt{\frac{\sum_{i=1}^{100}(y_i-\bar{y})^2}{100}}=6$

$\therefore \quad \sum_{i=1}^{100}\left(x_i-\bar{x}\right)^2=2500$ **and** $\sum_{i=1}^{100}\left(y_i-\bar{y}\right)^2=3600$

and it is also given that

$\sum_{i=1}^{100}\left(x_i-\bar{x}\right)\left(y_i-\bar{y}\right)=600$

$\because \quad a^2+b^2-2ab=(a-b)^2$

$\sum_{i=1}^{100}\left(x_i-\bar{x}\right)^2+\sum_{i=1}^{100}\left(y_i-\bar{y}\right)^2-\sum_{i=1}^{100}\left(x_i-\bar{x}\right)\left(y_i-\bar{y}\right)$

$=\sum_{i=1}^{100}\left(\left(x_i-\bar{x}\right)-\left(y_i-\bar{y}\right)\right)^2$

$\Rightarrow \quad \sum_{i=1}^{100}\left[\left(x_i-\bar{x}\right)-\left(y_i-\bar{y}\right)\right]^2=2500+3600-1200$

$\Rightarrow \quad \sum_{i=1}^{100}\left[\left(x_i-y_i\right)-\left(\bar{x}-\bar{y}\right)\right]^2=4900$

$\Rightarrow \sqrt{\frac{\sum_{i=1}^{100}\left[\left(x_i-y_i\right)-\left(\bar{x}-\bar{y}\right)\right]^2}{100}}=7$

Q.73 Answer:

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Solution:

Coordinate of vertices of triangle

$A(2,3), B(2,4), C(4,3)$

$\therefore \quad AB = 1, BC = \sqrt{5}, CA = 2$

So, $\triangle ABC$ is right angle triangle where right angle at A that is orthocentre also.

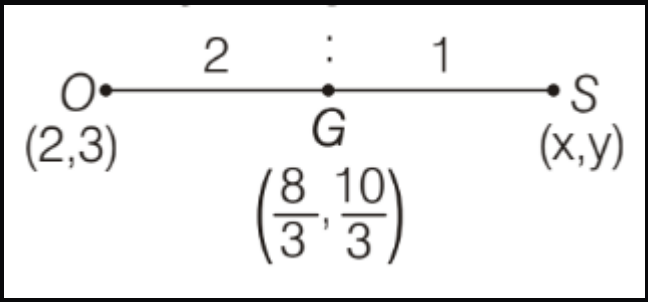
Coordinate of orthocentre is $O(2,3)$.

Coordinates of centroid

$$= \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

$$G = \left(\frac{8}{3}, \frac{10}{3}\right)$$

G divide the line joining O and S in the ratio $2:1$



$\frac{8}{3} = \frac{2x+2}{3} \Rightarrow x = 3$

$\frac{10}{3} = \frac{2y+3}{3} \Rightarrow y = \frac{7}{2}$

$\Rightarrow S\left(3, \frac{7}{2}\right)$

So, $AO^2 + 9BG^2 + 4CS^2$

$AO^2 = (2-2)^2 + (3-3)^2 = 0 \Rightarrow AO^2 = 0$

$BG^2 = \left(2 - \frac{8}{3}\right)^2 + \left(4 - \frac{10}{3}\right)^2 = \frac{8}{9} \Rightarrow 9BG^2 = 8$

$$CS^2 = (4-3)^2 + \left(3 - \frac{7}{2}\right)^2 = \frac{5}{4} \Rightarrow 4CS^2 = 5$$

$\therefore AO^2 + 9BG^2 + 4CS^2 = 13.$

Q.74 Answer:

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Solution:

For greatest term in expansion:

$T_{r+1} > T_r$

$\Rightarrow {}^9C_r 2^r(|-3x|)^{9-r} > {}^9C_{r-1}2^{r-1}(|-3x|)^{9-(r-1)}$

$\Rightarrow \dfrac{9!}{r!(9-r)!}2 \geq \dfrac{9!}{(r-1)!(9-r+1)!}3x$

$\Rightarrow \dfrac{2}{r} \geq \dfrac{3}{9-r+1}\{\because x=1\}$

$\Rightarrow 20-2r \geq 3r \Rightarrow r \geq 4$

$\therefore r = 4$

$T_5 = {}^9C_4(2)^4[-3]^5 = \dfrac{9!}{4! \times 5!}(2)^4 \cdot 3^5$

$= \dfrac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 6} \times 2^4 \times 3^5 = 3^7 \cdot 2^5 \cdot 7$

$\therefore \alpha = 7, \beta = 5, \gamma = 1, \delta = 0$

$\Rightarrow \alpha + \beta + \gamma + \delta = 7+5+1+0 = 13$

Q.75 Answer:

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Solution:

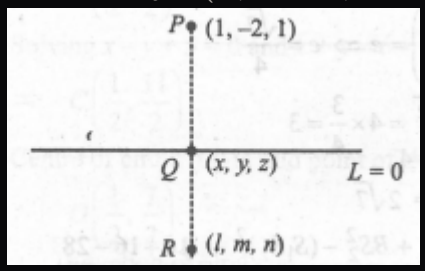
Equation of line passing through $B(1,1,2)$ **and** $C(2,2,1)$.

$L : \frac{x-1}{1} = \frac{y-1}{1} = \frac{z-2}{1} \leqslant k$

$\therefore$ **D.R. of parallel line to given line is** $(1,1,-1)$

$x = k+1, y = k+1, z = -k+2$

D.R. of PQ $: (k, k+3, 1-k)$



$\therefore \quad 1(k) + 1(k+3) - 1(1-k) = 0$

$\Rightarrow \quad 2k + 3 - 1 + k = 0 \Rightarrow 3k + 2 = 0$

$\Rightarrow \quad k = \dfrac{-2}{3}$

$\therefore \quad x = \dfrac{1}{3}, y = \dfrac{1}{3}, z = \dfrac{8}{3}$

Q **is mid point of** PR

$\Rightarrow \frac{1}{3} = \frac{1+l}{2} \Rightarrow l = \frac{-1}{3}$

$\Rightarrow \dfrac{1}{3} = \dfrac{-2+m}{2} \Rightarrow m = \dfrac{8}{3}$

$\Rightarrow \quad \dfrac{8}{3} = \dfrac{1+n}{2} \Rightarrow n = \dfrac{13}{3}$

$\therefore \quad l^2 + m^2 + n^2 = \dfrac{1}{9} + \dfrac{64}{9} + \dfrac{169}{9} = \dfrac{234}{9} = 26.$